INFORMATION AND INDUSTRY DYNAMICS

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Abstract

This paper develops a model of industry dynamics where firms compete to acquire customers over time by disseminating information about themselves in the presence of random shocks to their efficiency. The properties of the model's stationary equilibrium are related to empirical regularities on firm and industry dynamics. As an application of the model, the effects of a decline in the cost of information dissemination on firm and industry dynamics are explored.

JEL D80, L11, L16, M37

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Information and Industry Dynamics*

Emin M. Dinlersoz AND Mehmet Yorukoglu

This paper develops a model of industry dynamics where firms compete to acquire customers over time by disseminating information about themselves in the presence of random shocks to their efficiency. The properties of the model's stationary equilibrium are related to empirical regularities on firm and industry dynamics. As an application of the model, the effects of a decline in the cost of information dissemination on firm and industry dynamics are explored. (*JEL* D80, L11, L16, M37)

A fundamental task for any firm is to build a customer base. Technologies used by firms to reach and inform consumers about their attributes are therefore essential in firm survival and growth. Improvements in these technologies over time have reduced firms' costs of reaching beyond their local markets and allowed them to expand. Transmission of commercial information to wider markets started with the first true mass medium: power-driven printing combined with mass mailing by rail. The arrival of radio, then television, and most recently the Internet, gradually increased the reach, scope, and effectiveness of information dissemination. These innovations helped successful firms compete with rivals in distant markets and overtake them. The emergence of major corporations and national chains in the U.S. during the 20th century can in part be attributed to the rapid decline in the cost of information dissemination.[1] The escalation of international trade and the rise of multi-national corporations also owe much to the declining cost of penetration into foreign consumer masses.[2]

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[1] Chandler and Cortada (2000) provide an excellent account of how new information technologies have transformed the U.S. economy.

[2] See Arkolakis (2009) for the role of market penetration costs in international trade.

This paper explores the effects of technological progress in information dissemination on firm and industry dynamics. When information about firms spreads more quickly and at lower cost, a highly productive firm can expand more rapidly, whereas an unfavorable shock to a firm's productivity may result in faster downsizing and exit. Growth of Internet-based firms like Amazon.com exemplifies how reaching larger consumer populations more easily can speed up firm growth and dynamics. Amazon.com's employment five years from its inception was achieved only gradually over several decades by major firms founded earlier in the 20th century.[3] Amazon.com spent heavily to inform consumers and expand its customer base, even though this strategy led, in part, to large initial losses.[4] Enduring losses to ensure growth and future profitability through an emphasis on information dissemination and consumer awareness was a common practice in the early days of Internet-based commerce. Internet-based firms used heavy public awareness campaigns, including print and television ads, to inform potential customers. While the experience of Internet-based retail industries was highly publicized, the improvements in information dissemination technologies have influenced the formation and the ultimate structure of many other industries.[5]

To understand the role of information dissemination in firm and industry dynamics, we consider a market for a homogenous good where a large number of firms spread information to a large number of consumers to gradually accumulate a customer base. Firm entry and exit are allowed, and firms must thrive among many rivals, in contrast to markets with a small number of firms and little firm turnover. Firms experience random changes in their efficiency reflected in their marginal costs. To survive and grow, a firm must build a customer base in the presence of this randomness. Consumer information is key in this firm growth process. Information dissemination is akin to informative advertising, the main purpose of which is to let consumers know the price and other firm attributes relevant for consumer decision-making.[6] Information dissemination is not limited to well-organized

[3] It took HP and Allstate about 25 years, IBM and Lowe's nearly 20 years, and Walmart about 15 years to reach a comparable size. See Luttmer (2008).

[4] See, e.g., Frey and Cook (2004).

[5] See Sutton (1991).

[6] There is no effort by firms to persuade consumers or induce loyalty, although such efforts can be accommodated. Empirically, informative advertising appears to be as important as other types of advertising. See

media advertising campaigns, and can include e-mails, catalogs, postcards, or coupons aimed at informing or updating consumers about a firm's attributes.

Heterogeneity among firms in the model with respect to their efficiency highlights an important aspect of the provision of information: the effect of the information released by more efficient firms on less efficient ones. This negative pecuniary externality matters for the functioning of markets under competitive information dissemination. Advertising technology and the distribution of efficiency together determine how easily firms penetrate the market. A low advertising cost leads to overlapping reach by firms and favors more efficient firms which offer lower prices, while a high advertising cost can shield inefficient firms from competition. Improvements in efficiency allow a firm to expand, and the larger a firm, the higher is its value and the lower is the efficiency level it can withstand without exit. A firm's customer base is therefore a valuable asset.

Consumers in the model are passive information filters who can live for several time periods and make forward-looking decisions. Because consumers are long-lived, how they store and discard information about firms needs to be specified. There is a large number of small (infinitesimal) consumers and advertisements reach consumers randomly. Therefore, a consumer who received an ad from a firm and purchased from it in the past does not necessarily receive another ad from that same firm in the future. Unless a firm's past customers recall the firm or have updated information about the firm, there is no persistence in the firm's customer base. Some consumer memory or information update is needed to link a firm's clientele over time. We consider a specific but intuitive process as a baseline: a consumer recalls only the most recent firm he purchased from, and unless that firm exits, he is updated of that firm's current payoff-relevant information through means other than random advertising. This process implies that a consumer values a long-term relationship with a firm. A consumer obtains more surplus from visiting a firm with a lower likelihood of exit, because the consumer is more likely to have information about that firm in the future, and hence, more likely to have the option of purchasing from that firm. Since larger firms are also less likely to exit in the model, firm size matters for a consumer's decision to patronize a firm.

Bagwell (2007) for a discussion.

The mechanism of building a clientele gradually through information dissemination is at work in varying degrees in many industries. The model represents markets where a large number of firms offer highly substitutable products and information about firms and prices is disseminated into various media. In such markets consumers are not aware of all firms and basic price information is valuable, as consumers are interested mainly in prices and terms of sale. Furthermore, consumers in the model value the option of being able to purchase repeatedly from a firm. The model can thus approximate the environment of certain traditional and Internet-based retailers, wholesalers, mail-order companies, long distance phone and certain subscription based services, credit or insurance providers, stock brokers, and some manufacturing industries.[7]

The model's stationary equilibrium admits a number of properties relevant for firm and industry level evidence. A distinguishing feature is a two-dimensional price distribution. More efficient firms charge lower prices, but size also matters. Larger firms command higher prices, as they can appropriate some of the extra consumer surplus their size advantage generates through an increase in the likelihood of survival. The model also has implications on the evolution of price and advertising along a firm's life-cycle. Younger and smaller firms charge lower prices to secure a customer base. As a firm grows, pricing becomes less aggressive, resulting in higher prices for older and larger firms. This effect emerges without any reputation or brand-name associated with firm size or age – it is a consequence of the assumption that a consumer recalls his most-recently-visited firm. More efficient firms advertise more, and advertising escalates as a firm grows. Older firms are larger because firm size is persistent and it takes time to accumulate a customer base. Customer accumulation process in the model offers a micro-representation of reduced-form goodwill or demand accumulation processes frequently used in models of firm-level demand dynamics. The model can also generate a positively-skewed firm size without assuming any skewness or persistence in efficiency. Small firms need to be more efficient on average to be viable, and efficient small firms exhibit high growth rates, which decline eventually as a firm grows.

[7] Lambin (1976) argues that "...Almost all advertising by wholesalers, retailers (department stores, supermarket chains, mail order firms), and industrial and financial firms (savings banks, insurance) is informative in a narrow sense because of the prominent place given to prices and terms of sales...".

Firm size and age are negatively associated with exit. Age alone does not contain sufficient information to assess firm dynamics. Conditional on age, a firm's value and growth depend on size. We relate the properties of the model to existing empirical evidence, and provide new evidence on certain predictions of the model.

Using the model, we also investigate the effects of a decline in the cost of information dissemination. The direct effect is a higher incentive for firms to advertise. There is also an indirect effect. As firms reach more and more consumers at lower cost, competition among firms intensifies. More efficient firms release a larger number of ads containing lower prices, compared to inefficient firms. Ads from inefficient firms are then less likely to turn into sales, leading such firms to increase prices, and even exit. A decline in information cost also alters the mean and standard deviation of firm growth rates. It can speed up the growth of efficient firms and hasten the demise of inefficient ones, amplifying inherent firm volatility. Numerical analysis demonstrates that the decline can lead to higher industry concentration, higher entry and exit rates, faster firm growth, higher volatility in firm size and value, and an increase in the relative importance of firm efficiency compared to firm size in determining firm value. We present evidence on some of these effects for certain industries which experienced significant diffusion of Internet-based information dissemination and commerce.

This paper integrates two main strands of literature. Informative advertising has been analyzed in a static framework with identical firms, notably by Butters (1977) and Stegeman (1991). To these models, we add firm heterogeneity as well as firm and consumer dynamics. The dynamic setup is related to models of industry dynamics, including Jovanovic (1982), Hopenhayn (1992), Ericson and Pakes (1995), and Fishman and Rob (2002). We introduce informative advertising to this class of models. In Hopenhayn (1992), firm dynamics are determined solely by an exogenous productivity process. Here, in addition to efficiency, the accumulated customer base of a firm influences firm dynamics, and persistent efficiency is not required to generate persistent firm dynamics. Fishman and Rob (2002) consider industry dynamics with consumer search costs. While search and advertising both generate mechanisms of allocating consumers to firms, their implications on pricing and firm dynamics can differ.[8] Asplund and Nocke (2006) analyze the role of market size in the dynamics of

[8]For instance, in Fishman and Rob (2002) homogenous search costs lead to no price dispersion in equi-

imperfectly competitive industries. This paper investigates instead the effect of changes in firms' market reach on firm dynamics, given market size. Doraszelski and Markovich (2007) study dynamics generated by informative and goodwill advertising in a duopoly, whereas our focus is on an industry with many firms and firm turnover. The setup also relates to models of firm dynamics with consumer switching costs, e.g. Klemperer and Beggs (1992). While there is no explicit switching cost in the model, the fact that a consumer can default to his most-recently-visited firm introduces some consumer inertia: customers of larger and more efficient firms have higher reservation surplus and are less likely to switch.

The rest of the paper is organized as follows. Section I introduces the model. Section II describes the stationary equilibrium. Section III discusses properties of the equilibrium. Section IV studies some effects of improvements in information dissemination technology. Section V concludes. Proofs are in Web Appendix.

I. The Model

Consider an industry with a large number of firms offering a homogenous good to a large number of consumers. The population of firms and consumers are each represented by a continuum. An individual firm or consumer is by construction *small*, and has no influence on industry aggregates by itself and takes them as given. Time is discrete, and is denoted by $t \geq 1$. Figure 1 gives the timing of events and decisions within a period. At the beginning of each period, a potential entrant decides whether to enter before learning its marginal cost during its first period in the industry. Incumbents then decide whether to stay or irreversibly exit after observing their marginal cost for the current period. At that time, some consumers randomly exit the market, while new ones enter to replace the exiting ones. Firms then send ads to consumers, who make their purchase decisions. Firms and consumers have the same discount rate $\beta \in (0, 1)$.

A. *Consumers*

There is a unit mass of ex-ante identical consumers who demand the good every period. Each consumer has a fixed probability of exit, $\lambda \in (0, 1)$, in any given period,

librium. Consumers never switch firms, and new consumers are allocated among existing firms equally – a process that assigns no role to firm characteristics in luring consumers.

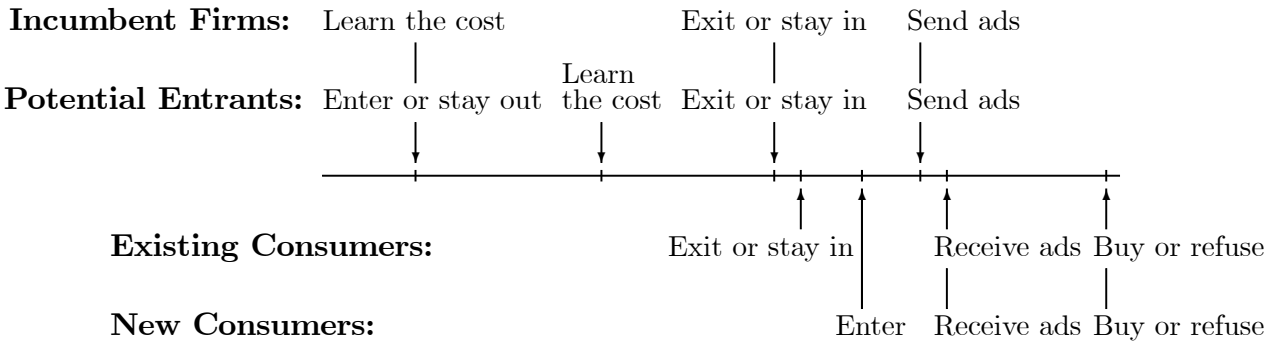


Figure 1: The timing of events and decisions within a period

implying an expected consumer life of $1/\lambda$ periods. Each period exiting consumers are replaced by new ones. Within any period, a consumer can purchase either zero or one unit of the good, resulting in a net surplus of

$$U(p) = \begin{cases} s - p, & \text{if the consumer buys at price } p, \\ 0, & \text{otherwise,} \end{cases}$$

where $s > 0$ is the gross surplus.

B. *Consumer Information and Memory*

Consumers receive ads randomly and independently. Each consumer is equally likely to receive an ad. Denote by $\Psi_t(i)$ the probability that a consumer receives $i \geq 0$ discrete ads in period t. Let I_t be the information flow to a consumer during period t. I_t contains information about the firms from which the consumer received an ad in period t, and about those firms visited in the past that provide the consumer with their updated (period t) information. I_t is either empty ($\varnothing$), implying no new information, or consists of $K > 0$ discrete elements. Each element is a vector of variables that describe the corresponding firm's attributes a consumer can use to identify a firm and evaluate the associated surplus.

Because consumers can survive for several periods, how a consumer stores and discards past information needs to be specified. If a consumer does not recall any of the firms he

was informed of in the past and has no information updates from those firms, there is no persistence in firm size. This result follows because the probability that a consumer receives an ad from the same firm in any two different periods is zero, as a firm's customers has negligible mass (measure zero) with respect to the total mass of consumers. There are several ways to specify consumer memory. At one extreme, a consumer recalls all past information (total recall). At the other, a consumer forgets the past (no recall). Let $\mathcal{M}_t$ denote a consumer's memory. No recall corresponds to $\mathcal{M}_t^{\varnothing} \equiv \varnothing$, and total recall to $\mathcal{M}_t^{\tau} \equiv \bigcup_{j=t-\tau}^{t-1} I_j$, where $\tau \geq 1$ is consumer age. Any arbitrary memory process can be represented as a subset of $\mathcal{M}_t^{\tau}$. For instance, one can define a k-period memory, $\mathcal{M}_t^{k} \equiv \bigcup_{j=t-k}^{t-1} I_j$, as one that contains all the past information flow from period $t-k$ up to period $t-1$, $k \leq \tau$. Another example is a memory that contains only the firms from which a purchase was made. A consumer's cumulative information, $\mathcal{I}_t$, is the collection of his memory and current information flow, i.e. $\mathcal{I}_t = \mathcal{M}_t \cup I_t$. A consumer makes his purchase decision based on $\mathcal{I}_t$.

Our baseline specification about consumer information and memory is a specific, but intuitive one: consumers do not recall any ads received or firms visited in the past, except that if a consumer purchased from a firm in the previous period and that firm did not exit, the consumer possesses the current period payoff-relevant information about that firm. The baseline corresponds to a no recall memory ($\mathcal{M}_t^{\varnothing}$) and to an information flow I_t that contains the information in the new ads received plus the period-t information about the most-recently-visited firm, if any. The baseline cumulative information process is thus $\mathcal{I}_t \equiv \mathcal{M}_t^{\varnothing} \cup I_t \equiv I_t$.

The baseline specification is motivated by the fact that a consumer becomes more familiar with the firm he purchased from in the previous period, compared to the firms he received an ad from but chose not to visit. A consumer may also more likely forget the information about firms visited in distant past. The most-recently-visited firm is thus likely the 'freshest' firm in consumer memory. The consumer may receive the current payoff-relevant information about this firm through means other than random ads. Lists of previous customers are frequently used by firms to update their former customers directly. Current catalogs or coupons may be sent directly to previous customers' addresses. The cost of advertising defined below can include the cost of sending such updates to consumers who purchased from the firm

in the previous period. The assumed information structure generates persistence in a firm's customer base. We discuss alternative information structures in Section III.F.

Next, consider the contents of I_t. As in static models of informative advertising, a firm's price matters for consumer surplus and is a part of each element in I_t. In a dynamic environment where consumers can purchase from a firm repeatedly, firm attributes that matter for firm survival are also pay-off relevant. The baseline memory process implies that the firm visited by a consumer in period t will be in I_{t+1}, unless that firm exits. Consumers therefore attach value on firm attributes, such as size, that matter for survival. We assume that firm size is revealed in an ad. Alternatively, proxies for firm size can be included in an ad, or firm size may become common knowledge over time once a firm grows beyond a certain size.[9] A consumer does not know a firm's efficiency.

C. *A Consumer's Dynamic Problem*

If $\mathcal{I}_t = \varnothing$, the consumer does not purchase in period t. Otherwise, the consumer purchases from the firm in $\mathcal{I}_t$ that provides the highest surplus. Because a consumer has the option of purchasing from the most-recently-visited firm, consumers are heterogeneous with respect to their reservation values, which change over time randomly as new information is received. Let $W_t(p, m^o)$ be a consumer's discounted expected surplus from a firm that charges price p during period t and that had $m^o \geq 0$ customers at the end of period $t-1$. We refer to m^o as the "size" of the firm at the beginning of period t. Let $X_t(p, m^o)$ be the probability of exit at the beginning of period t. Also, let $W_t^{\varnothing}$ be the discounted expected surplus of a consumer with $\mathcal{I}_t = \varnothing$. Note that $W_t(p, m^o) \geq W_t^{\varnothing} > 0$, i.e. being informed of some firm brings at least as much surplus as having no information. We can write

$$
\begin{aligned}
W_t(p, m^o) \;=\; & U(p) \qquad (1)\\
& +\beta(1-\lambda)\left\{\Psi_{t+1}(0)[(1-X_{t+1}(p,m^o))E\left[W_{t+1}(p',m')\right] + X_{t+1}(p,m^o)W_{t+1}^{\varnothing}\right]\\
& +\sum_{i=1}^{\infty}\Psi_{t+1}(i)\left[(1-X_{t+1}(p,m^o))E\left[\int \max\{W, W_{t+1}(p',m')\}dG_{t+1}^{i}(W)\right]\right.\\
& \left.+\; X_{t+1}(p,m^o)\int W dG_{t+1}^{i}(W)\right]\Big\}.
\end{aligned}
$$

[9] Firm age, which is positively associated with firm size in equilibrium, can proxy for firm size.

In (1), G_t^i is the *c.d.f.* of the first-order statistic of consumer surplus among $i > 0$ ads received by the consumer in period t. Because ads from different firms arrive independently, one can write $G_t^i(W) = [G_t(W)]^i$, where $G_t(W)$ is the probability that an ad is from a firm that offers a surplus of at most W. The expectations in (1) are taken with respect to all feasible next period pairs (p', m') for the firm conditional on survival in period $t+1$.

The first term on the right hand side of (1) is the period t surplus. The second term is the expected discounted future surplus if no ad is received. The second term has two components: the future surplus, $E[W_{t+1}(p', m')]$, if the consumer's current firm stays in the market, and the future surplus if that firm exits

$$W_{t+1}^{\varnothing} = \beta(1-\lambda)\left\{\Psi_{t+2}(0)W_{t+2}^{\varnothing} + \sum_{i=1}^{\infty}\Psi_{t+2}(i)\int W dG_{t+2}^{i}(W)\right\}. \tag{2}$$

In (2), the surplus in period $t+1$ is zero, and the following period's surplus depends on the number of ads received. If no ad is received, the surplus is $W_{t+2}^{\varnothing}$, otherwise the consumer chooses the firm that yields the highest surplus. The third and fourth terms in (1) together give the expected future surplus from receiving one or more ads. The third term applies when the consumer's current firm stays in the market. In this case, $\mathcal{I}_{t+1}$ has $i+1$ elements corresponding to i new ads and the updated information about the most recently visited firm. The consumer chooses the firm that offers the highest surplus. Similarly, the fourth term represents the case where the current firm exits and the consumer has i new ads. A consumer chooses the firm in $\mathcal{I}_t$ that provides the highest surplus

$$\overline{W}_t(\mathcal{I}_t) = \begin{cases} \max\limits_{i=1,\ldots,K} W_t(p_i, m_i^o), & \text{if } |\mathcal{I}_t| = K \geq 1, \\ W_t^{\varnothing}, & \text{if } \mathcal{I}_t = \varnothing. \end{cases} \tag{3}$$

Consider now the probability that a consumer purchases from firm (p, m^o), conditional on having information about that firm. This probability is also the demand the firm faces from a consumer informed of the firm. Each such consumer can be of two types: one who purchased from the firm in the previous period, and the other who did not. Let the demand functions for these two types of consumers be $z_t(p, m^o)$ and $\omega_t(p, m^o)$, respectively. Because the latter type may have purchased in the previous period from another firm that provides higher current surplus than firm (p, m^o), firm (p, m^o) faces less demand from the latter type.

Lemma 1 *$z_t(p, m^o) \geq \omega_t(p, m^o)$, and price-elasticities satisfy $\varepsilon_{w_t}(p, m^o) \geq \varepsilon_{z_t}(p, m^o)$.*

The function $z_t(p, m^o)$ can be more explicitly written as

$$z_t(p, m^o) = \sum_{i=0}^{\infty} \Psi_t(i) G_t^i(W_t(p, m^o)), \tag{4}$$

where the ith term is the probability of sale to a consumer who has i ads from firms other than firm (p, m^o). Furthermore, using the proof of Lemma 1,

$$\omega_t(p, m^o) = [\theta_t + (1 - \theta_t)\alpha_t(p, m^o)] z_t(p, m^o). \tag{5}$$

where θ_t is the measure of consumers with no information from other firms, and $\alpha_t(p, m^o)$ is the probability that the firm a consumer purchased from in period $t-1$ provides less surplus than firm (p, m^o). Lemma 1 suggests that a firm would like to charge a higher price to its previous-period customers compared to the new ones. We assume away any price discrimination. A firm's price therefore reflects the composition of its customers, as discussed below.

D. *Firms*

Firms pay a one-time sunk cost of $\kappa > 0$ upon entry. A firm's marginal cost, c, is a realization of a continuous random variable with *c.d.f.* $H(c)$ and differentiable density $h(c)$ over $[\underline{c}, \infty)$ with $s > \underline{c} > 0$, such that $H(c) < 1$ for all $c \in [\underline{c}, \infty)$ and $\lim_{c \to \infty} H(c) = 1$. Marginal costs are generated from H *independently* (over time) and *identically.*[10] An *i.i.d.* distribution isolates the role of customer accumulation process in generating persistence in firm size. If marginal cost exhibits persistence, then both the persistence in the marginal cost and the persistence in a firm's customer base contribute to persistence in firm size. Because $H(c) < 1$ for all $c \in [\underline{c}, \infty)$, there is a positive probability that a firm's marginal cost is larger than the highest sustainable marginal cost without exit. As a result, there is always some exit, consistent with continuing turnover even in mature industries. An incumbent firm incurs a

[10] We assume that the realized frequency distribution of marginal cost in a period coincides with the actual distribution H. Certain conditions are needed for this law of large numbers with a continuum of agents (see, e.g., Feldman and Gilles (1985)). Following Feldman and Gilles (1985, p. 28), one solution is to assume that firms closely located in the marginal cost space have correlated cost shocks within a period.

fixed cost of $f > 0$ per period, which can be avoided only if the firm exits. Since all firms have the same fixed cost, a firm with a lower marginal cost than another is "more efficient".

E. *Information Technology*

Firms disseminate information using a common advertising technology, $\Phi(a)$, which gives the total cost of sending $a \in [0, \infty)$ ads, assumed to be equivalent to the total cost of reaching a distinct consumers.

Assumption 1 *(i)* Φ *is twice differentiable and* $\Phi' > 0$ *and* $\Phi'' > 0$ *on* $(0, \infty)$, *(ii)* $\Phi(0) = 0$, *and (iii)* $\lim_{a \to 0} \Phi'(a) = 0$.

Assumption 1(i) captures the basic empirical regularity of diminishing returns to advertising, as documented in Sutton (1991). From an individual firm's perspective, there is a very large, inexhaustible pool of consumers. The advertising cost can be viewed as the cost of ensuring that each ad is received by a distinct consumer, and no consumer receives more than one ad from a firm. If the advertising technology exhibits constant or increasing returns to scale, a firm can reach an arbitrarily large number of consumers and grow without bound. Convexity in advertising cost prevents such an outcome. As Grossman and Shapiro (1984) point out, increasing marginal cost of advertising can stem from media saturation, overlapping media, or the existence of different predispositions to view ads in the consumer population. As the amount of ads sent increases, it becomes increasingly costly for the firm to find an additional consumer who has not yet received an ad from the firm. In other words, deeper penetration to consumer population requires increasingly higher marginal cost of marketing.

Assumptions 1(ii) and 1(iii) are made for simplicity and ensure that all surviving firms advertise, preventing the possibility that a firm coasts without advertising. Any unavoidable fixed cost of advertising can be included in the fixed cost, f. A positive fixed cost of advertising or a positive marginal cost around zero can lead to no advertising for some firms in a period.[11]

[11] One class of advertising technologies which satisfy Assumption 1 is $\Phi(a) = \chi a^{\varphi}$, $\varphi > 1, \chi > 0$. One class which violates Assumption 1(iii), but satisfies Assumptions 1(i)-(ii), is $\Phi(a) = \chi(e^{\varphi a} - 1)$, $\varphi, \chi > 0$.

F. *A Firm's Dynamic Problem*

The period profit of a firm with size m^o and marginal cost c, when it charges price p and sends a ads, is

$$\Pi_t(p, a; m^o, c) = m_t(p, a; m^o)(p - c) - \Phi(a) - f, \tag{6}$$

where m_t is the firm's residual demand, or the end-of-period number of customers, given by

$$m_t(p, a; m^o) = m^o(1 - \lambda)z_t(p, m^o) + a\omega_t(p, m^o). \tag{7}$$

Π_t is strictly increasing in m_t, strictly concave in a, and strictly decreasing in c and f. The law of motion (7) can also be interpreted as capital accumulation subject to stochastic depreciation and investment. In this interpretation, m^o is the firm's capital stock at the beginning of period t. During period t, a fraction $\lambda + (1 - \lambda)(1 - z_t(p, m^o))$ of this stock depreciates, where λ is the industry-wide depreciation rate common to all firms and $(1 - z_t(p, m^o))$ is the firm-specific depreciation rate. A fraction $\omega_t(p, m^o)$ of investment a turns into new capital, where $\omega_t(p, m^o)$ is the firm-specific investment success rate. The total cost of investment is $\Phi(a)$. Equation (7) also offers a micro-representation of goodwill or demand accumulation processes frequently used in models of firm-level demand dynamics, which are typically specified in a reduced form.

Let the pair (m^o, c) be a firm's *type*. A firm's value is

$$V_t(m^o, c) = \max\left\{0, \max_{p,a}\left[\Pi_t(p, a; m^o, c) + \beta \int V_{t+1}(m_t, c')h(c')dc'\right]\right\}. \tag{8}$$

The first term inside the curly brackets is the value of exit normalized to zero, and the second term is the maximized sum of the current profit and the discounted expected future profit. A firm is assumed to stay in the industry when indifferent. Entry is unrestricted. Each period a mass $M_t \geq 0$ of new firms enter the industry. An entrant starts off with zero customers. Free entry implies

$$\int_{\underline{c}}^{\infty} V_t(0, c)h(c)dc \leq \kappa, \tag{9}$$

which holds with equality when $M_t > 0$.

The state of the industry in period t is described by a measure μ_t over the set of firm types (m^o, c). For any Borel set $S \subset \mathbb{R}^+ \times [\underline{c}, \infty)$, $\mu_t(S)$ gives the mass of firms with types in S. Given an initial measure of firms μ_o, μ_t results from entry, exit and changes in incumbents' sizes and efficiencies up to time t. Let $B_t(p, m^o)$ denote the set of firm types that provide a consumer a surplus of at most $W_t(p, m^o)$. The probability that an ad comes from some firm type in $B_t(p, m^o)$ is

$$Q_t(p, m^o; \mu_t) = \frac{1}{A_t} \left(\int_{B_t(p,m^o)} a_t(m, c) d\mu_t(m, c) \right), \tag{10}$$

where $a_t(m, c)$ is the number of ads sent by a type (m, c) firm, and $A_t = \int_{(m,c)} a_t(m, c) d\mu_t(m, c)$ is the total number of ads. Noting that $Q_t(p, m^o; \mu_t) \equiv G_t(W_t(p, m^o; \mu_t))$, the individual demand function in (4) can be expressed as $z_t(p, m^o; \mu_t) = \sum_{i=0}^{\infty} \Psi_t(i) Q_t^i(p, m^o; \mu_t)$. Similarly, $\omega_t(p, m^o)$ can also be related to μ_t using (5).

II. Stationary Equilibrium

Let (*) identify variables and functions in equilibrium.

Definition 1 (Stationary equilibrium) *A stationary equilibrium consists of pricing and advertising policies $p^*(m^o, c)$ and $a^*(m^o, c)$, a cumulative distribution G^* of consumer surplus across ads, a measure μ^* over firm types, an entry mass $M^* \geq 0$, and an exit rule $x^*(c) \geq 0$, such that, in any given period of time: (i) Firms maximize their profits by adopting policies $p^*(m^o, c)$ and $a^*(m^o, c)$, (ii) Consumers maximize their surplus, (iii) Firm types (m^o, c) with $m^o < x^*(c)$ exit, (iv) M^* satisfies the free entry condition (9), (v) G^* is consistent with (i) through (iv), and (vi) μ^* is consistent with (i) through (v).*

In the case of free advertising, i.e. $\Phi(a) = 0$ for $a \geq 0$, every consumer has information about all firms. If there is a positive mass of firms, price competition implies that firms offering the highest surplus charge a price that leaves them with zero period profit gross of the fixed cost, while all other firms make no sales. This polar case is ruled out by Assumption

1. The rest of the analysis focuses on equilibria with strictly positive advertising cost for $a \in (0, \infty)$, where a positive mass of firms in the industry compete with each other.[12]

Assuming for now a stationary equilibrium with positive advertising, entry, and exit exists, we characterize the equilibrium. Let $z^*(p, m^o)$ and $\omega^*(p, m^o)$ be the individual demand functions in equilibrium, and $\widetilde{z}^*(m^o, c)$ and $\widetilde{\omega}^*(m^o, c)$ be the demands evaluated at the optimal pricing policy. The firm's customer base is given by $m^*(m^o, c) = m^*(p^*, a^*; m^o)$. Following the theory of stochastic dynamic programming in Stokey and Lucas (1993), there exists a unique continuous and bounded time-invariant function V^* that is a fixed point of the operator defined by (8).

Let $W^*(p, m^o)$ be the surplus from visiting firm (m^o, c), and let $W^{\varnothing *}$ be the surplus of an uninformed consumer. W^* and $W^{\varnothing *}$ are the fixed-points of the operators defined by (1) and (2), respectively. Once again, the dynamic programming techniques in Stokey and Lucas (1993) ensure the existence and uniqueness of continuous and bounded functions W^* and $W^{\varnothing *}$. There exists then a unique bounded function $\overline{W}^*(\mathcal{I})$ that gives the maximum surplus associated with any $\mathcal{I}$. The individual demand functions, firm value, and consumer surplus satisfy the following properties.

Proposition 1 *(i) $z^*(p, m^o)$ and $\omega^*(p, m^o)$ are strictly decreasing in p, (ii) $z^*(p, m^o)$ and $\omega^*(p, m^o)$ are strictly increasing in m^o, and (iii) $V^*(m^o, c)$ is strictly decreasing in c, and strictly increasing in m^o, and (iv) $W^*(p, m^o)$ is strictly decreasing in p, and strictly increasing in m^o.*

Proposition 1(iii) implies that exit probability strictly increases as the marginal cost increases, and the minimum size, $x^*(c)$, a firm with marginal cost c can sustain without exit is a strictly increasing function of c. A firm exits the industry when $m^o < x^*(c)$.[13]

Consider next the pricing and advertising policies. Given the price distribution, the firm type that provides the lowest surplus to a consumer makes a sale when it is the only firm

[12] In the other polar case of very high advertising costs, no firm advertises. Given other parameters of the model one can find an advertising technology such that $\lim_{a \to 0} \Phi'(a) > N$ for any real number $N > 0$, i.e., the marginal cost of advertising around zero exceeds the marginal benefit from advertising even for the firm type that provides the highest surplus to a consumer.

[13] Alternatively, a firm exits if c is greater than the threshold $c^*(m^o)$.

a consumer is informed of. Hence, that firm type charges its monopoly price.[14] For all other firms, the optimal price, $p^*(m^o, c)$, is an interior solution that satisfies the first order condition

$$m^*(p, a; m^o) + \frac{\partial m^*(p, a; m^o)}{\partial p}\left[(p - c) + \beta\frac{\partial}{\partial m^*}\int V^*(m^*, c')h(c')dc'\right] = 0. \qquad (11)$$

A firm chooses its price considering the dynamic effect of this choice. This effect is represented by the second term inside the brackets in (11), which is the change in the firm's continuation value due to a small change in its customer base m^* at the beginning of the next period. Denote that term by R^*. The relative markup is then $(p^* - c + R^*)/p^* = 1/\varepsilon^*(p^*, m^o)$, where ε^* is the weighted average of the elasticities of demand by the firm's previous-period and new customers

$$\varepsilon^*(p^*, m^o) = \frac{m^o(1-\lambda)z^*(p^*, m^o)}{m^*(p^*, a; m^o)}\varepsilon_{z^*}(p^*, m^o) + \frac{a\omega^*(p^*, m^o)}{m^*(p^*, a; m^o)}\varepsilon_{\omega^*}(p^*, m^o). \qquad (12)$$

In other words, with no price discrimination a firm's price reflects the composition of its customers.

More efficient firms charge lower prices compared to less efficient ones of the same size. Conditional on efficiency, a larger firm offers more surplus because of its lower probability of exit, which implies that customers of this firm are more likely to have the option of being able to purchase from this firm in the future, compared to a smaller firm. This additional surplus allows larger firms to charge higher prices. Therefore, size and efficiency have opposite effects on price. Competition among firms prevents firms from appropriating the full consumer surplus. The firm type that offers the lowest surplus leaves a consumer a surplus of $W^{\varnothing *}$, making the consumer indifferent between buying versus refusing. Other firm types leave consumers a surplus strictly higher than $W^{\varnothing *}$. However, if the connection between firm size and consumer surplus is absent, as in the case when exit is independent of firm size (e.g. exit with an exogenously fixed probability), or when consumers forget all past information, then a firm's price does not depend on its size.

Proposition 2 *(i) There exists a unique pricing policy, $p^*(m^o, c)$, which is strictly increasing in m^o and strictly decreasing in c, (ii) If exit does not depend on m^o, then pricing policy is also independent of m^o.*

[14] Monopoly price can exceed a consumer's period gross surplus s, because W^* can exceed s.

In equilibrium, some firms can set a price below marginal cost, sacrificing current profits for future profitability. Below-cost pricing can be observed for young and small firms with high marginal cost.

Assumption 1 ensures that firms advertise every period. The advertising policy, $a^*(m^o, c)$, is defined implicitly by the first order condition

$$\frac{\partial m^*(p, a; m^o)}{\partial a} \left[(p - c) + \beta \frac{\partial}{\partial m^*} \int V^*(m^*, c') h(c') dc' \right] - \Phi'(a) = 0. \tag{13}$$

As in the choice of price, a change in current period's advertising level affects the firm's continuation value. The advertising policy is an increasing function of a firm's efficiency and size, because the expected profit from marginal ad is higher for firms with larger clientele and higher efficiency.

Proposition 3 *There exists a unique advertising policy, $a^*(m^o, c)$, which is strictly increasing in m^o and strictly decreasing in c.*

Denote the sales of a firm by $r^*(m^o, c) = m^*(m^o, c)p^*(m^o, c)$.

Proposition 4 *A firm's demand, $m^*(m^o, c)$, and sales, $r^*(m^o, c)$, are strictly increasing in m^o and strictly decreasing in c.*

Because a firm's end-of-period number of customers, $m^*(m^o, c)$ is strictly increasing in previous-period size m^o, there is persistence in firm size. This persistence drives some properties of equilibrium. The results so far were obtained under the assumption that there exists a stationary equilibrium with positive advertising, entry, and exit. Under certain conditions a unique such equilibrium exists – see Theorem 1 in Web Appendix.

III. Properties of the Stationary Equilibrium

A. *Firm Size, Efficiency, and Advertising*

In the model a firm's current advertising expenditures are positively associated with the firm's current efficiency and previous-period size (Proposition 3). A positive relation between firm size and advertising expenditures is empirically well-established.[15] Little evidence

[15] See, e.g., Sutton (1991) and Bagwell (2007).

exists, however, on efficiency-advertising relationship. We present further evidence on both relationships using data from US Census Bureau's Census of Manufactures (CM), a quinquennial census of manufacturing establishments. Following the model, we relate a firm's advertising expenditures to its previous size and current efficiency. Advertising expenditures is the sum of advertising cost reported by all establishments of a firm in 1997 CM, including the cost of advertising by a firm's auxiliary and administrative establishments. We allocate such advertising expenditures to an industry based on the share of a firm's total value of shipments classified in that industry.

Previous firm size is measured in alternative ways: total value of shipments and total employment in 1992 CM, all aggregated to firm level from establishment level. For firms that entered between 1992 and 1997 CM, 1992 firm size measures are set to zero. We do not observe the 1997 advertising expenditures and efficiency for firms that exited between 1992 and 1997. However, we control for this selection in our analysis. The measure of efficiency in the model is marginal cost, which cannot be recovered easily from the available data.[16] Instead, we use a firm's 1997 revenue-based total factor productivity (TFP) as a proxy for efficiency.[17]

We first use OLS estimation to explore the relationship between advertising expenditures, size, and efficiency. We account for selection due to exit by using Heckman's (1979) two-step procedure.[18] Because many firms report zero advertising, we also use a Tobit model to control for any potential censoring. As discussed earlier, zero advertising can be obtained in the model by introducing a fixed cost of advertising (modifying Assumption 1(ii)), or by modifying Assumption 1(iii), either of which leads to a corner solution, and therefore,

[16] For a small subset of establishments in CM, physical output and some cost measures are reported. We utilize this information to calculate an average variable cost in our investigation of prices.

[17] See Web Appendix. The TFP measure accounts for the fact that some firms produce in more than one 4-digit industry.

[18] The first-step is a probit model of selection (survival) probability, where in addition to the variables used in the second step, we use variables that are likely correlated with survival but not necessarily with advertising. Our choice of such variables is limited in CM. The variables we used are an indicator for whether a firm had positive exports in the sample year, primary industry specialization ratio, and primary product specialization ratio. We also used firm age (measured in years by the age of the firm's oldest establishment) and the results were similar.

potential left-censoring in the data.[19] Both specifications I and II in Table 1 indicate a significant positive association between size measures and advertising expenditures. The specifications also suggest a positive relationship between TFP and advertising expenditures. The inverse Mills Ratio in the second-step OLS regression consistently has a positive sign, indicating that unobserved factors that make survival more likely are positively related to advertising. One concern is that the previous size measure in Table 1 pertains to five years prior to 1997 CM. Five years may be too long depending on persistence in firm size. For robustness, we also use the size measures in the 1996 version of US Census Bureau's Annual Survey of Manufactures (ASM). ASM has an annual frequency, but pertains only to a sample of large establishments. The results for the ASM sample turn out to be similar, as shown in Table A1 in Web Appendix.

B. *Firm Size, Age, and Survival*

Empirical studies (e.g. Evans (1987) and Dunne, Roberts, and Samuelson (1989)) have consistently found that larger firms have a higher probability of survival, older firms are larger on average, and size increases with age in a first order stochastic sense. In the model, the first finding arises because firm value is increasing in size. Stationary equilibrium is also consistent with the latter two findings.

Proposition 5 (*i*) *Firm size increases with firm age in a first order stochastic sense,* (*ii*) *average firm size increases with firm age, and* (*iii*) *older firms have lower exit rates on average.*

Proposition 5 is obtained under no persistence in efficiency (*i.i.d.* shocks to marginal cost). Instead, the persistence in a firm's clientele causes the positive relation between firm age and size, and the negative relation between firm age and exit probability, consistent with higher hazard rates for younger firms (e.g. Dunne, Roberts, and Samuelson (1989)).

C. *Firm Size, Efficiency, and Price*

[19] We use a data-driven censoring threshold, equal to the minimum of the observed advertising expenditures. See Carson and Sun (2007) for details. The results were also similar when we used an OLS regression after omitting firms with zero advertising expenditures.

The model has implications on a firm's pricing along its life-cycle. Smaller firms have lower prices conditional on efficiency, and more efficient firms also have lower prices conditional on size (Proposition 2). Because entrants start off with zero customers, the most efficient entrant type has the lowest price. Younger and smaller firms tend to charge lower prices to build a customer base. As a firm grows, its pricing tends to become less aggressive. Firm efficiency decreases in a first order stochastic sense as firm size increases. Smaller firms are therefore on average more efficient than larger firms. A similar result is obtained by Fishman and Rob (2002) in a model of industry dynamics with consumer search.

Some recent evidence supports the predictions of the model on prices. Foster, Haltiwanger, and Syverson (2008) find that younger businesses tend to have higher physical productivity and lower prices compared to older incumbents in manufacturing industries with highly homogeneous products. Large firms in Internet retailing, such as Amazon.com and Barnes and Noble.com, also do not charge the lowest prices when compared to their smaller competitors (see, e.g., Brynjolfsson and Smith (2000) and Smith and Brynjolfsson (2001)). We provide further evidence on the relationship between a firm's price, size, and efficiency. For a subset of establishments in CM, measures for physical quantities shipped are available for their products at the 7-digit SIC code level. From this subset, we use data for 11 manufacturing industries consisting of highly homogenous products and establishments that are highly specialized in these products.[20] Because the model focuses on homogenous goods and firms specialized in a single product, its implications on price can be more cleanly tested using this dataset. Price is defined as the value of shipments divided by the quantity shipped by all establishments of a firm. Quantity serves as an alternative measure of firm size and is also used to compute a firm-level quantity-based (physical) TFP, following Foster, Haltiwanger, and Syverson (2008). As an alternative measure of efficiency, we also use an average variable cost measure (AVC).[21]

Following the model, we relate a firm's price in 1992 CM to its previous size, measured

[20] The products are boxes, bread, carbon black, coffee, concrete, flooring, gasoline, block ice, processed ice, plywood, and sugar. For details on this dataset, see Foster, Haltiwanger, and Syverson (2008). We thank the authors for providing their dataset.

[21] To calculate AVC, we first multiply the revenue-share of the 7-digit product by the firm's total labor and material costs, and then divide the resulting number by the quantity shipped for that 7-digit product.

by its value of shipments, employment, or quantity shipped in 1987 CM, and to its efficiency, as measured by its quantity-based TFP or AVC in 1992 CM.[22] We focus only on the firms that appear in both censuses. Various specifications of OLS estimation in Table 2 indicate a positive association between price and firm size, and a negative association between price and TFP. The estimated elasticities for size measures vary across specifications, with the quantity measure having the lowest elasticity. The estimated elasticities for efficiency measures are higher in absolute value than those for size measures. The elasticity estimates for efficiency measures are also much higher in absolute value when quantity-based TFP is used, compared to AVC. Overall, the estimated coefficients in Table 2 lend some support to Proposition 2. However, the samples represent only a very small fraction of all products and firms in U.S., and the generality of the results remains to be seen.

D. *Firm Size Distribution*

Empirical studies indicate a positively skewed firm size distribution with a decreasing firm size density at least at the right tail. Pareto distribution appears to provide a good fit for firm size greater than some threshold (see, e.g., Cabral and Mata (2003)). The model can generate a decreasing firm size density and a skewed firm size distribution. A firm needs a sufficiently long streak of favorable cost shocks to become large. Such streaks become increasingly rare as the ultimate size to be achieved increases, implying a decreasing firm size density at least for large firm size levels, and a positively skewed firm size distribution.

Let $f^*(m)$ be the stationary firm size density over the support $[\underline{m}^*, \overline{m}^*]$, where $\underline{m}^* > 0$ and $\overline{m}^* < \infty$.[23] Let $c^*(m, m^o) \in [\underline{c}, \infty)$ be the largest cost shock a firm can withstand to

[22] The sample is much smaller for 1997, because no quantity information was collected for concrete in 1997. Therefore, we did not use the 1997 CM sample. Unlike in the case of samples used in analyzing advertising expenditures in Section III.A, the ASM samples are small for the products considered in price analysis, so we do not use 1991 ASM size measures.

[23] Positive fixed cost implies $\underline{m}^* > 0$. The maximum size $\overline{m}_t^*$ at age t is attained by an entrant which receives the minimum marginal cost draw $\underline{c}$ every period after entry: $\overline{m}_t^* = a^*(\overline{m}_{t-1}^*, \underline{c})\widetilde{\omega}^*(\overline{m}_{t-1}^*, \underline{c}) + (1-\lambda)Z_{t-2}^* a^*(\overline{m}_{t-2}^*, \underline{c})\widetilde{\omega}^*(\overline{m}_{t-2}^*, \underline{c}) + ... + (1-\lambda)^{t-1} Z_0^* a^*(0, \underline{c})\widetilde{\omega}^*(0, \underline{c})$, where $Z_j^* = \prod_{\tau=1}^{t-(j+1)} \widetilde{z}^*(\overline{m}_{t-\tau}^*, \underline{c})$. By Propositions 1 and 3, $a^*(\overline{m}_t^*, \underline{c})\widetilde{\omega}^*(\overline{m}_t^*, \underline{c})$ is a strictly increasing function of t. Furthermore, for any t, $Z_t^* a^*(\overline{m}_t^*, \underline{c})\widetilde{\omega}^*(\overline{m}_t^*, \underline{c}) \leq a^*(\overline{m}_t^*, \underline{c}) < \infty$, because $\widetilde{\omega}^*(\overline{m}_t^*, \underline{c}) \leq 1$ and $Z_t^* \leq 1$. Therefore, $\overline{m}_t^* \leq$

achieve a size of at least m at the end of the period, starting from a previous period size of $m^o \geq 0$. The function $c^*(m, m^o)$ is strictly decreasing in m, as a higher size next period requires a lower current cost. The conditional probability that next period size is at most m is

$$P^*(m|m^o) = \begin{cases} 0 & \text{if } c^*(m, m^o) < \underline{c}, \\ 1 - H(c^*(m, m^o)) & \text{if } c^*(m, m^o) \in [\underline{c}, c^*(m^o)). \end{cases} \tag{14}$$

The stationary firm size density $f^*(m)$ can be expressed as a fixed point of the operator that relates current firm size density to that of the next-period

$$f^*(m) = \int_{\underline{m}^*}^{\overline{m}^*} f^*(m|m^o) f^*(m^o) dm^o + f^*(m|0) P^*(0), \tag{15}$$

where $f^*(m|m^o) = \frac{dP^*(m|m^o)}{dm}$ is the firm size density conditional on previous period size and $P^*(0)$ is the fraction of firms that are new entrants. For firm size to exhibit a non-increasing density throughout its entire support, one stringent sufficient condition is $f^{*\prime}(m|m^o) \leq 0$ for $m^o \geq 0$, or equivalently, using (14),

$$\frac{h'(c^*(m, m^o))}{h(c^*(m, m^o))} \geq -\frac{\dfrac{\partial^2 c^*(m, m^o)}{\partial m^2}}{\left[\dfrac{\partial c^*(m, m^o)}{\partial m}\right]^2}. \tag{16}$$

When the right hand side of (16) is positive, (16) holds for density functions for the marginal cost distributions with sufficiently high positive first derivative ($h' > 0$). For a given marginal cost distribution, the right hand side of (16) is positive when the advertising cost increases steeply enough so that $\dfrac{\partial^2 c^*(m, m^o)}{\partial m^2} \leq 0$. If the right hand side of (16) is non-positive, (16) can hold for densities such as uniform, exponential, and Pareto, for which $h' \leq 0$.

E. *Firm size, growth, and Gibrat's Law*

Growth rate for a type (m^o, c) firm conditional on survival ($c \leq c^*(m^o)$) is

$$g^*(m^o, c) = \frac{m^*(m^o, c) - m^o}{m^o} = (1 - \lambda)\widetilde{z}^*(m^o, c) + \frac{a^*(m^o, c)\widetilde{\omega}^*(m^o, c)}{m^o} - 1, \tag{17}$$

$a^*(\overline{m}^*_{t-1}, \underline{c})\frac{[1-(1-\lambda)^t]}{\lambda}$. As a result, $\overline{m}^* = \lim_{t\to\infty} \overline{m}^*_t \leq \lim_{t\to\infty} a^*(\overline{m}^*_{t-1}, \underline{c})\frac{[1-(1-\lambda)^t]}{\lambda} = \frac{a^*(\overline{m}^*, \underline{c})}{\lambda} < \infty$ by continuity of $a^*(m^o, c)$ in m^o.

and the expected growth rate conditional on size and survival is $\overline{g}^*(m^o) = E_c[g^*(m^o, c)|m^o, c \leq c^*(m^o)]$. Conditional on size and survival, more efficient firms have higher growth rates, because $g^*(m^o, c)$ is strictly decreasing in c. Because $g^*(0, c) = +\infty$ for $c \leq c^*(0)$, it holds that $\overline{g}^*(0) = +\infty$, implying that new entrants have the highest expected growth rates.[24] As firm size increases, the expected growth rate must eventually become negative. To see this claim, note that from (17) and the facts that $\widetilde{z}^*(m^o, c) \leq 1$ and $\widetilde{\omega}^*(m^o, c) \leq 1$, one can write $\lim_{m^o \to \overline{m}^*} \overline{g}^*(m^o) \leq -\lambda + E_c\left[\frac{a^*(\overline{m}^*, c)}{\overline{m}^*}\right] \leq \lambda\left(E_c\left[\frac{a^*(\overline{m}^*, c)}{a^*(\overline{m}^*, \underline{c})}\right] - 1\right) < 0$, because $\overline{m}^* = \frac{a^*(\overline{m}^*, \underline{c})}{\lambda}$ and $a^*(\overline{m}^*, c) < a^*(\overline{m}^*, \underline{c})$ for $c > \underline{c}$. The sign of the expected growth rate over the support of firm size depends on how the number of new customers acquired, $a^*\widetilde{\omega}^*$, changes with firm size. If $a^*\widetilde{\omega}^*$ is a sufficiently concave function of size, then the expected growth rate can start to decline even at relatively small firm size levels. In the special case of full consumer turnover ($\lambda = 1$), current size is independent of previous-period size. Both $g^*(m^o, c)$ and $\overline{g}^*(m^o)$ are then strictly decreasing in m^o. The dependence of current firm size on previous size becomes weaker as λ increases.

In summary, Gibrat's Law of independence of firm growth and size, as incorporated in some models of firm growth, does not necessarily emerge in the model. Gibrat's Law does not appear to hold for entrants and established firms tracked for a five-year or longer time period (e.g. Lotti, Santarelli, and Vivarelli (2003)). Empirical work by Evans (1987) and Dunne, Roberts, and Samuelson (1989) also points to a general failure of Gibrat's Law in US manufacturing.

F. *Robustness*

Consider first a longer consumer memory than the baseline case. Suppose a consumer of age $\tau \geq 1$ recalls all the firms he was ever informed of – total recall, $\mathcal{M}_t^\tau$. The consumer needs to assess expected period-t surplus from each firm in I_{t-k}, $k = 0, 1, ..., \tau$. The consumer can infer the size of each firm at the beginning of period $t - k + 1$ from period $t - k$ size and price information. Among the firms in I_{t-k}, the largest firm at the beginning of period $t - k + 1$ provides the highest expected surplus at time t because cost shocks are *i.i.d.*.

[24] This result follows because $\widetilde{z}^*(0, c)$, $\widetilde{\omega}^*(0, c)$, and $a^*(0, c)$ are strictly positive, as there is always a positive mass of uninformed consumers when advertising is not free.

Remembering only the largest firm at the beginning of period $t-k+1$ is sufficient for assessing the highest possible surplus associated with I_{t-k}. The consumer can compare the expected period-t surplus from these largest firms using each firm's size and the number of periods, k, the consumer has known about each of these firms. Because consumers now have more information compared to the baseline case, firms effectively face more competition. Pricing is also more aggressive especially among larger and more efficient firms which are more likely to be recalled by consumers, leading to changes in the incentives to advertise by both efficient and inefficient firms, and also in entry and exit rates. Other memory processes, all subsets of total recall memory $\mathcal{M}_t^\tau$, can also be considered. For instance, a consumer can recall randomly one of the firms visited in the past, or the largest firm ever visited. The model's general properties can hold under a variety of such specifications.

Consider next relaxing the assumption that a consumer is updated of the current-period payoff-relevant information about the most-recently-visited firm. Suppose that a consumer recalls only the period $t-1$ size and price of the most-recently-visited firm, but receives no update about its period t price. The consumer can then assess the expected surplus from the firm based on period $t-1$ information. This modification preserves persistence in firm size; however, a consumer's problem becomes more complicated, as discussed in the preceding paragraph.

The assumption that firm size is revealed in an ad can also be modified. If firm size is not revealed, consumers must take into account possible future firm sizes in forming expectations about the surplus offered by a firm. An alternative is to assume that firm size is common knowledge, or more plausibly, becomes so after a firm grows to a certain size. Another alternative is to assume that firm age, which is positively correlated with size, is revealed in an ad.

It is also possible to break the firm size-consumer surplus link that works through repeat purchase probability. Suppose that the discount factor β is set to zero for consumers.[25] A consumer's problem then becomes static. If consumers are allowed to recall their most-recently-visited firms and have updated information from those firms, firm size still exhibits persistence, but consumers do not place any value on repeat purchase probability and firm

[25] We thank an anonymous referee for this suggestion.

size, as they do not care about the future. In this case, price is the only payoff-relevant variable for consumers.

Assumption 1 places little restriction on the technology other than convexity. Advertising cost, however, can in some cases be non-convex over $a \in (0, \infty)$. Such cases can arise, for instance, if an ad is shared, e.g. through word-of-mouth or some consumer network, by several consumers at no additional cost to the firm. If the advertising cost is not strictly convex over the interior of its domain, certain features of advertising policy, such as uniqueness and monotonicity, may no longer hold. A fixed cost of advertising can be introduced, or the slope of the advertising technology around zero can be made positive. In such cases, in a given period some firm types do not advertise, but the main results continue to hold.

Finally, suppose that marginal cost follows a persistent stochastic process.[26] Such persistence reinforces many features of the equilibrium. A firm's size, advertising, and pricing becomes more persistent over time, and inequality in firm value and size across different levels of efficiency can increase.

IV. The Role of Information Cost

Consider two otherwise identical economies where it is cheaper to spread information on the margin in the second economy: $\Phi_1'(a) > \Phi_2'(a)$ for all $a > 0$, implying $\Phi_1(a) > \Phi_2(a)$ for all $a > 0$ by Assumption 1. The direct effect of a decline in the cost of information dissemination is an incentive for all firms to advertise more. However, there is also an indirect effect, which changes the expected return to an ad – the first term on the left hand side of (13). The indirect effect depends not only on a firm's efficiency but also on the collective advertising made by all firms that offer more surplus than the firm. When advertising outlays of larger and more efficient firms increase, the mass of consumers who receive ads offering higher surplus becomes larger. Less efficient and smaller firms' ads are then less likely to turn into a sale. The value of such firms can decrease, leading to more exit. A small yet efficient

[26] The possibility of efficiency signalling through prices now arises. If there exists a separating equilibrium, consumers can infer a firm's type correctly. On the other hand, a pooling equilibrium where a positive mass of firms with different types charge the same price cannot exist. If there was pooling at a price, any of the firms charging the common price can lower its price and steal a positive mass of consumers from other firms with the same price, ending up with a discrete gain.

firm can have a higher value in the low-information-cost economy, because it can build a large clientele more easily and grow faster. On the other hand, even a large firm can fail more easily in response to a large unfavorable shock. Therefore, a decline in information cost changes both the mean and standard deviation of firm growth rates. Because the decline affects firms with different efficiencies disparately, inequality in firm value and size can also change. The exact nature of these effects depend on the advertising technology and the distribution of marginal cost. We examine numerically how the equilibrium responds to changes in the information dissemination cost using certain configurations for the model's parameters. The algorithm used for computing the stationary equilibrium is in Web Appendix.

A. *Baseline Specification*

The distribution H of marginal cost is chosen to be uniform (over $[1, 5]$) to demonstrate that the model can deliver a skewed firm size distribution without any skewness in efficiency.[27] The advertising cost is a strictly convex function in accordance with Assumption 1: $\Phi(a) = \chi a^{\varphi}$, $\chi > 0$ and $\varphi > 1$. Our baseline specification is $\chi = 1$ and $\varphi = 2$. This advertising technology generates positive advertising under both the baseline and other specifications used below (see Web Appendix). The discount factor β is 0.95. The entry cost and the fixed cost are $\kappa = 0.1$ and $f = 0.5$, respectively. Finally, consumer gross period surplus is $s = 4$ and the probability of consumer exit is $\lambda = 0.1$, implying an expected consumer life-time of 10 periods. In the simulations below, 20 discrete grid points are used for the support of the marginal cost. Firm size is represented by 200 discrete grid points.

B. *Scale of Information Cost*

We first change χ from 1 to 2, scaling up the marginal and total cost of advertising by a constant factor at all levels of advertising. The firm types that exit are the ones that lie below the exit schedules in Figure 2. The exit region is larger in the low-information-cost economy. Figure 3 presents the average market value of firms as a function of marginal cost.

[27] While the upper bound of the support of H can be arbitrarily large, in practice what is needed is that the upper bound be large enough so that any firm with a cost draw close to that upper bound exits with certainty.

Firms have higher average value in the high-information-cost economy at all marginal cost levels. Firm size has a lower effect on average firm value in the low-information-cost economy, as shown in Figure 4, where, for each size grid, the market values are averaged over marginal cost levels for surviving firms at that size grid. Larger firms have higher value in the high-information-cost economy where it is more difficult to grow, but also easier to remain large. In contrast, in the low-information-cost economy, a small yet efficient firm attracts consumers more easily, and a large firm that experiences a bad shock loses its customers more easily. Firm size-value profile is steeper in the high-information-cost economy, implying that value effect of an increase in size is higher at all size levels. Figure 5 shows a positive association between firm age and size, as implied by Proposition 5. Figure 5 also demonstrates that firm growth is gradual, and when the information cost is lower, firms grow faster to a given size. In the high-information-cost economy, firms are much older on average for a given size.

Figure 6 plots volatility, as measured by the standard deviation of the next-period-size *conditional on* current size *and* survival, i.e. $Vol(m^o) = [V(m^*(m^o, c)|m^o, m^o \geq x^*(c))]^{1/2}$. In both economies, larger firms exhibit lower volatility. In the low-information-cost economy volatility is higher for all size levels. Figure 7 shows industry concentration, defined as the fraction of sales accounted by firms larger than a given size percentile. Sales are much more concentrated in the low-information-cost economy. For instance, in the high-information-cost economy the top decile accounts for $\sim$30% of sales, compared to $\sim$45% in the low-information-cost economy. The density of firm size is plotted in Figure 8. The size distribution is positively skewed and downward sloping at the right tail, in line with empirical evidence. In the low-information-cost economy the size distribution is more skewed compared to the high-information-cost economy (2.6 versus 0.98).

C. *Convexity of Information Cost*

In this exercise, the value of the convexity parameter is changed from $\phi = 2$ to $\phi = 3$. This modification implies a higher marginal cost of advertising for $a > 2/3$ and a steeper rise in marginal cost for $a > 1/3$, compared to the baseline case. Web Appendix contains the figures for this exercise. As shown in Figure 9, changing the convexity of information cost does not affect the exit region substantially, compared to the re-scaling

exercise. Nevertheless, for the less-convex-cost case, the exit region is larger. The relationship between firm value and firm size is shown in Figure 10. Firms have lower average market value at larger size levels under more convex cost, as it is more costly to sustain larger sizes. For the same reason, average market value of firms also increases less steeply as firm size increases.

The firm size-age relationship is in Figure 11. As in Figure 5, average firm size increases with age. In the more convex case, however, it takes less time on average for firms to grow, at least up to a certain size range. Smaller firms have a cost advantage because of the higher convexity in advertising technology that leads to a much steeper increase in the cost of sending ads as the advertising outlays increase beyond a certain level. Above a certain size, average firm age is higher for the more convex case, as a firm has to experience on average a longer streak of favorable shocks to achieve a given size. The volatility of firm size is plotted in Figure 12. Similar to Figure 6, less convex advertising cost generates higher volatility for all firm size levels. Size distributions in Figure 13 exhibit positive skewness and downward slope as in the case of the re-scaling exercise and skewness decreases (from 2.6 to 2.1) as convexity increases.

D. *Some Empirical Evidence*

The numerical analysis suggests that a lower cost of information dissemination can increase entry and exit, industry concentration, and firm size volatility. Campbell, Lettau, Malkiel and Xu (2001) find an increase in firm stock return volatility for publicly-traded firms in U.S. between 1962 and 1997. Employment and sales volatility of publicly held firms also increased, as documented by Comin and Philippon (2005). Davis, Haltiwanger, Jarmin and Miranda (2006) find, using more comprehensive data, that while employment volatility increased among publicly traded firms, it declined among privately held firms between 1976 and 2002. The model studied here does not have an explicit mechanism for employment volatility. The volatility of sales is a more appropriate measure from the model's perspective. Sales can be more volatile than employment in response to changes in the competitive environment. We present some evidence on sales volatility below.

There is also some prior work on the effects of information technologies on the evolution

of industry aggregates. Eckard (1988) found that between 1963 and 1977, a period during which TV advertising diffused substantially, prices fell and both output and concentration increased in a sample of 4-digit manufacturing industries where TV advertising was adopted by firms. Mueller and Rogers (1980) observed that TV advertising was positively related to sales concentration in a sample of 4-digit manufacturing industries. Baye and Morgan (2003) found that average industry concentration across nearly 5,000 online product-markets increased between 2000 and 2002, as Internet-based commerce diffused.

To provide further evidence, we identified certain retail and service industries for which the implications of the model are likely relevant.[28] The industries in Table 3 experienced significant diffusion of Internet-based information dissemination and commerce since early 1990's and share a number of common features relevant for the model's environment. Many firms in these industries release information about their prices on the Internet through their websites, ads, listings in search engines, and e-mails to existing and new customers. These industries have also experienced significant diffusion of Internet-based information dissemination and commerce. While reliable e-commerce penetration rates (e-commerce sales as a percent of total sales) are not available at the 4-digit SIC level, the penetration rates for certain product categories sold by electronic shopping and mail-order houses are provided by US Census Bureau.[29] Some of these products are offered by the industries in Table 3. In 1999, the first year the statistics were released, penetration rates were as follows: Books and Magazines (45.2%), Electronics and Appliances (17.7%), and Music and Videos (18.0%). In 2007, the corresponding rates rose to 61%, 74.1%, and 74.0%. The industries in Table 3 have also been the focus of recent research regarding the effect of easier information exchange between firms and consumers made possible by the Internet.[30]

We document the evolution of each industry from 1992 onwards. In 1992, Internet-based commerce was small, and by 2006, the last year we calculate a statistic for, it had grown substantially. We study the evolution of the number of firms, skewness in firm size, and

[28] See Web Appendix for industry definitions.

[29] See `http://www.census.gov/econ/estats/index.html`

[30] See, e.g., Brynjolfsson and Smith (2000) for books and CD's, Brown and Goolsbee (2001) for life insurance, and Baye, Morgan, and Scholten (2004) for electronic equipment. For Life Insurance, Brown and Goolsbee (2001) found lower prices as a result of online dissemination of price information.

sales concentration using Hirschman-Herfindahl index (HHI), and 4-firm concentration ratio (CR4). The numerical analysis suggests an increase in the relative sizes of larger firms compared to smaller firms, and an increase in the fraction of sales accounted by larger firms compared to that by smaller firms, as the cost of information dissemination declines. We thus expect both the inequality in sales across firms and the share of large firms in sales to increase. HHI captures the former effect, whereas CR4 captures the latter. Concentration and skewness measures, and the number of firms are based on the 1992, 1997, and 2002 economic censuses.[31] We report both the nationwide concentration and the average concentration across Metropolitan Statistical Areas (MSA), which are proxies for local markets.[32]

Finally, to obtain estimates of sales volatility, suppose that a surviving firm's sales r_t^* follows the process

$$r_t^* = \gamma_t(r_{t-1}^*) + \sigma_t(r_{t-1}^*)\varepsilon_t, \tag{18}$$

where the error term is assumed to satisfy $E[\varepsilon_t|r_{t-1}^*] = 0$ and $V(\varepsilon_t|r_{t-1}^*) = 1$, implying that $\gamma_t(r_{t-1}^*) = E[r_t^*|r_{t-1}^*]$ and $\sigma_t(r_{t-1}^*) = V(r_t^*|r_{t-1}^*)$.[33] The specification in (18) is an ARCH process, and $\sigma_t(\cdot)$ is the volatility function. The volatility measure, $Vol(m_o)$, reported in the numerical analysis is monotonically related to this function. Given an estimator $\widehat{\sigma}_t(\cdot)$ of $\sigma_t(\cdot)$, average volatility across firms in industry i at time t can be estimated as $\widehat{\sigma}_{it} = \frac{1}{N_{it}}\sum_{j=1}^{N_{it}}\widehat{\sigma}_t(r_{it-1})$, where $j = 1, ..., N_{it}$ indexes surviving firms. The estimator $\widehat{\sigma}_t(\cdot)$ we use is a non-parametric one due to Fan and Yao (1998). First, a local linear estimator $\widehat{\gamma}_t(\cdot)$ is constructed for $\gamma_t(\cdot)$ in (18). The squared residuals from this estimation are then used to obtain a local linear estimator $\widehat{\sigma}_t^2(\cdot)$, the square root of which yields $\widehat{\sigma}_t(\cdot)$.[34]

[31] At the time of this study, the final versions of the 2007 economic censuses were not available. Our calculations based on preliminary versions of the 2007 economic censuses were consistent with the patterns found using these three censuses.

[32] MSA definitions are as of June 30, 1999.

[33] Sales is given by $r_t^* = p_t^* m_t^* = p_t^*\left(m^o(1-\lambda)z_t^* + a_t^*\omega_t^*\right)$. Noting that $m^o = \frac{r_{t-1}^*}{p_{t-1}^*}$, the expected sales $E[r_t^*]$ is a non-linear function of r_{t-1}^* given by $\gamma^*(r_{t-1}^*) = E_c\left[p_t^*(\frac{r_{t-1}^*}{p_{t-1}^*}, c)\left(\frac{r_{t-1}^*}{p_{t-1}^*}(1-\lambda)z_t^*(\frac{r_{t-1}^*}{p_{t-1}^*}, c) + a_t^*(\frac{r_{t-1}^*}{p_{t-1}^*}, c)\omega_t^*(\frac{r_{t-1}^*}{p_{t-1}^*}, c)\right)\right]$.

[34] Specifically, let $\widehat{\gamma}(r_{t-1}^*) = \widehat{a}$ where $(\widehat{a}, \widehat{b}) = \arg\min_{a,b}\sum_{j=1}^{N_{it}}[r_{jt}^* - a - b(r_{jt-1}^* - r_{t-1}^*)]^2 K\left(\frac{r_{jt-1}^* - r_{t-1}^*}{h_1}\right)$. Also let $\widehat{\varepsilon}_t^2 = (r_t^* - \widehat{\gamma}(r_{t-1}^*))^2$. The estimator is then $\widehat{\sigma}_t^2(r_{t-1}^*) = \widehat{\alpha}$, where $(\widehat{\alpha}, \widehat{\beta}) = \arg\min_{\alpha,\beta}\sum_{j=1}^{N_{it}}[\widehat{\varepsilon}_{jt}^2 - \alpha - \beta(r_{jt-1}^* - r_{t-1}^*)]^2 W\left(\frac{r_{jt-1}^* - r_{t-1}^*}{h_2}\right)$. The functions $K(\cdot)$ and $W(\cdot)$ are kernels and h_1 and h_2 are the bandwidths. See Fan and Yao (1998) for further details.

The annual volatility estimates $\widehat{\sigma}_{it}$ are obtained from Business Register (BR), which provides annual sales for firms. BR is US Census Bureau's main annual source of employment and payroll measures. Reliable sales measures in the BR start in 1994, so we use 1995 as the first year of volatility estimates, because we require lagged sales.[35] We report the time-average of $\widehat{\sigma}_{it}$ for two periods, 1995-2000 and 2001-2007. The former is an initial stage of the diffusion of Internet-based commerce, and the latter is a more progressed one.

As shown in Table 3, concentration measures increased for all industries between 1997 and 2002 during which Internet-based commerce diffused faster. There was an increase in concentration between 1992 and 1997 with exceptions for New Car Dealers, Travel Agencies, and Life Insurance, but the increase was generally less compared to the 1997-2002 period. Table 4 indicates a net exit between 1992 and 2002 in all industries except for New Car Dealers. For Tape, CD & Record Stores and Travel Agencies, the number of firms increased between 1992 and 1997, before declining in 2002 below 1992 levels. Overall, the changes in the number of firms across the industries are consistent with escalated exit emerging in the numerical analysis, considering the fact that the market size for these industries also likely grew as population increased during 1992-2002. Skewness of firm size increased between 1992 and 2002 for almost all industries. The increase was more pronounced during 1997-2002, and skewness actually declined between 1992 and 1997 for New Car Dealers, Travel Agencies, and Life Insurance. In all cases, sales volatility was higher for the 2001-2007 period, compared to the 1995-2000 period. Overall, the findings, while limited to a small sample of industries, give some support to the patterns suggested by the numerical analysis.

V. Conclusion

Among many factors that have influenced firm and industry dynamics over time, this paper focused on one: the declining cost of information dissemination by firms to consumers. Introduction of an information dissemination technology to a dynamic industry framework has implications for price dynamics over a firm's life-cycle, the distribution of prices over

[35] Some firms are late in reporting their receipts, which are included in the following two years' data. Following Spletzer (1998), we made adjustments to sales data to ensure as much coverage of firms as possible. In some cases, a firm was part of a larger enterprise, and it did not have separately reported sales. We allocated the enterprise sales to the firm based on the firm's share of either enterprise payroll or employment.

firm size and efficiency, firm size distribution, and firm turnover and growth patterns. The model was used to analyze the effects of a decline in information costs. Easier spread of information can speed up the demise of inefficient firms, enhance the dominance of more efficient ones, and lead to higher concentration. Numerical analysis suggests that lower cost of information dissemination can alter the effects of random shocks to firms' efficiency, leading to changes in firms' market value and size distribution, and the mean and variance of firm growth rates. The evolution of some retail and service industries that experienced diffusion of Internet-based commerce is consistent with some of these changes.

The framework can accommodate extensions. Broader marketing activities by firms can be introduced, such as investments that increase the likelihood that ads turn into sale. Both the reach and effectiveness of information dissemination on industry dynamics can then be studied. The implications of limited consumer memory and bounded information processing capability on competition and firm dynamics can also be investigated. A limited consumer memory can encourage a firm to invest in helping consumers recall the firm, potentially leading to expensive and repetitive ads by firms in an effort to become one of the few firms in consumer memory. The model can also be used to explore the role of information costs in determining the responsiveness of prices and markups to expansions versus recessions.

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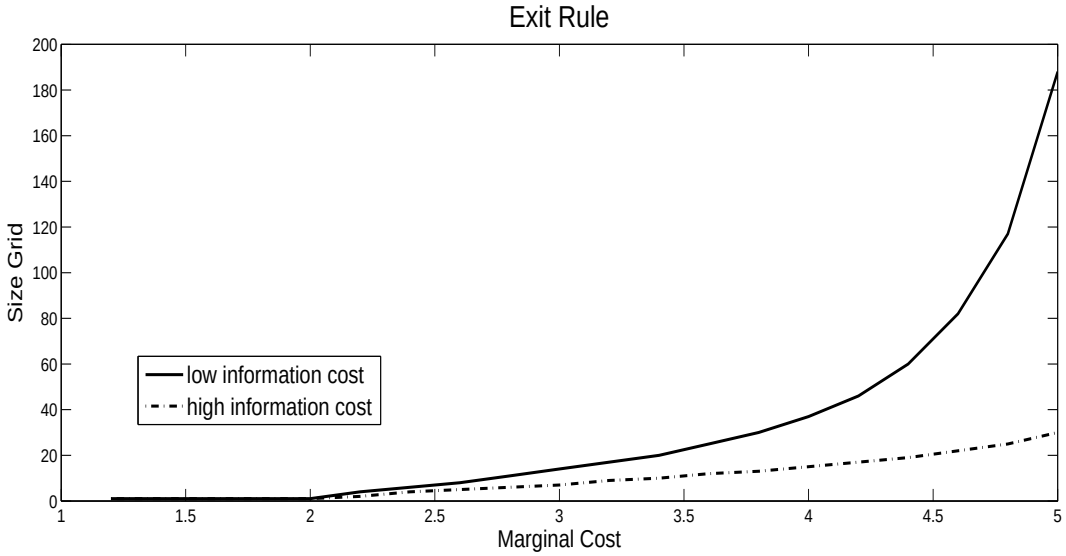


Figure 2: Scale of information cost: Exit rules

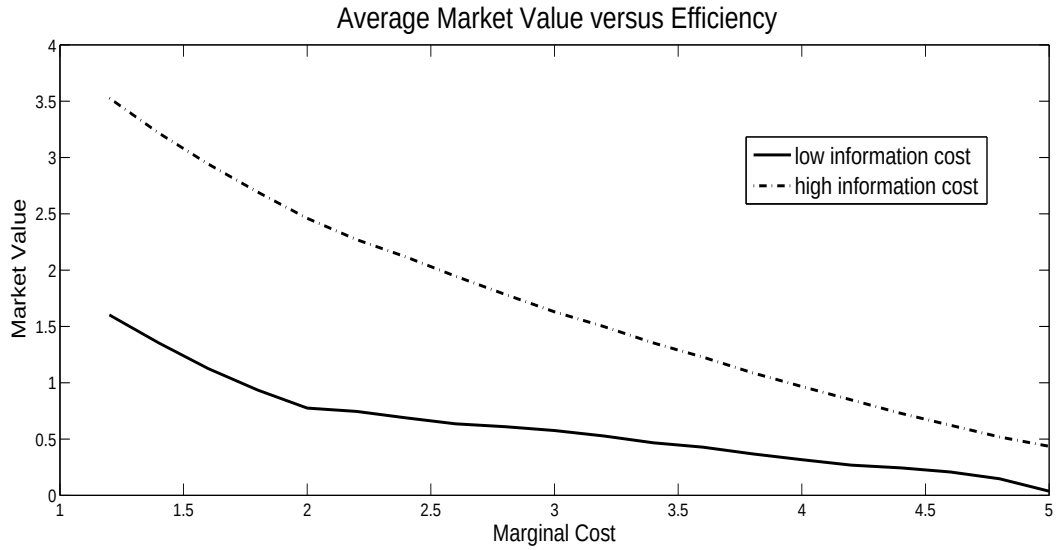


Figure 3: Scale of information cost: Firm efficiency-value relationship

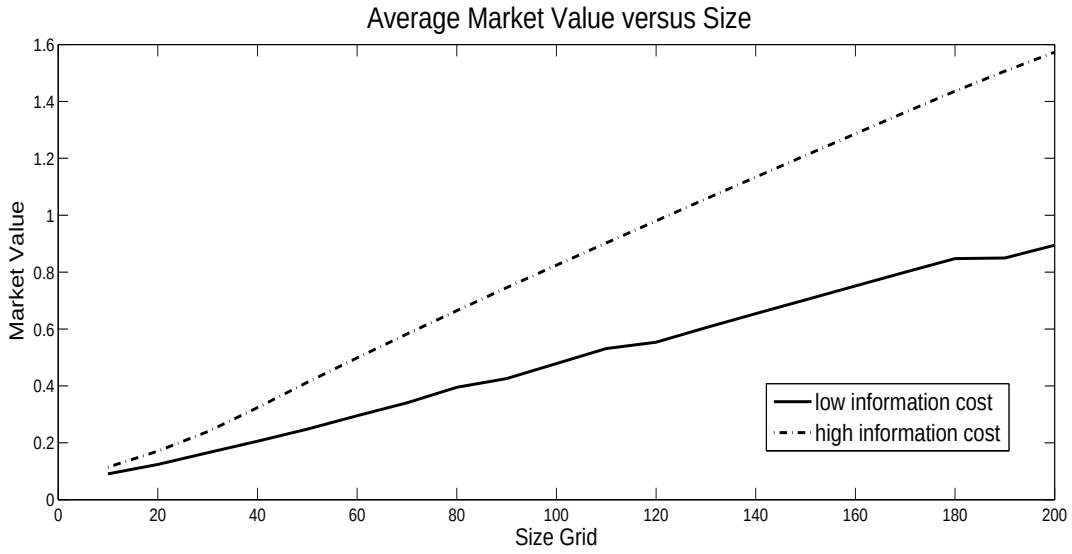


Figure 4: Scale of information cost: Firm size-value relationship ("0"=smallest)

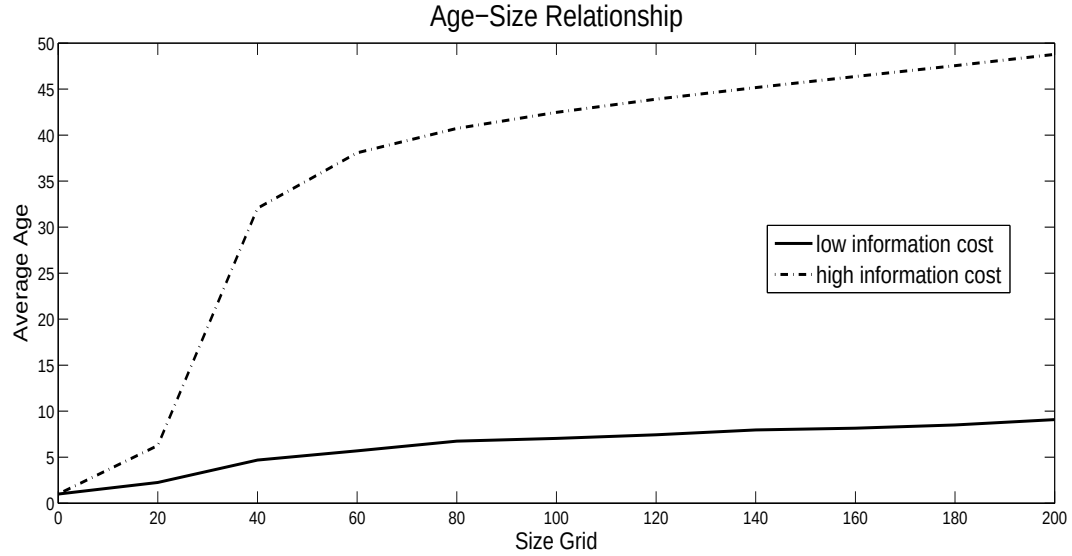


Figure 5: Scale of information cost: Firm age-size relationship ("0"=smallest)

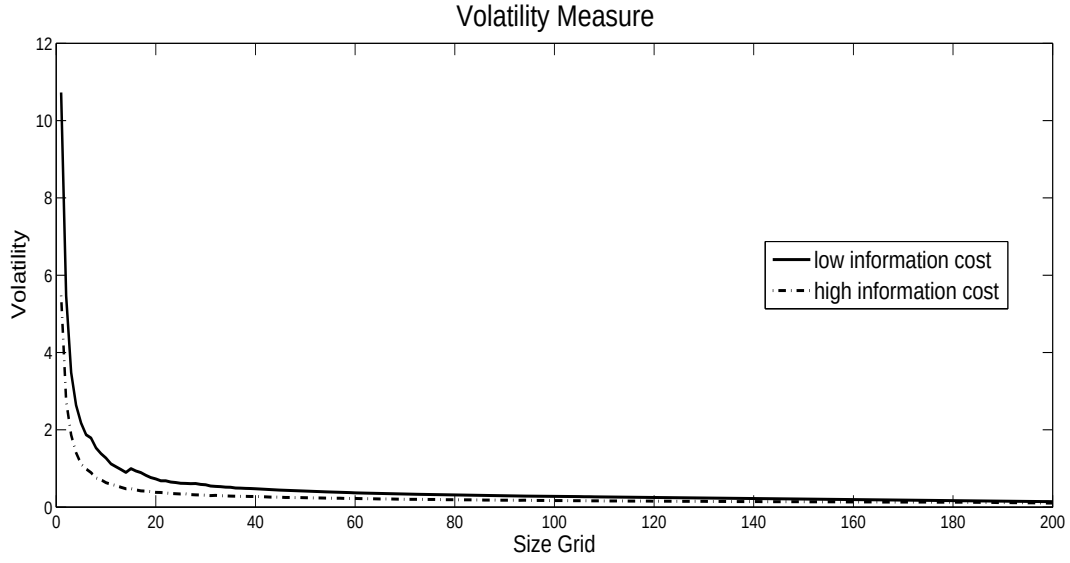


Figure 6: Scale of information cost: Firm size-volatility relationship ($Vol(m^o)$) ("0"=smallest)

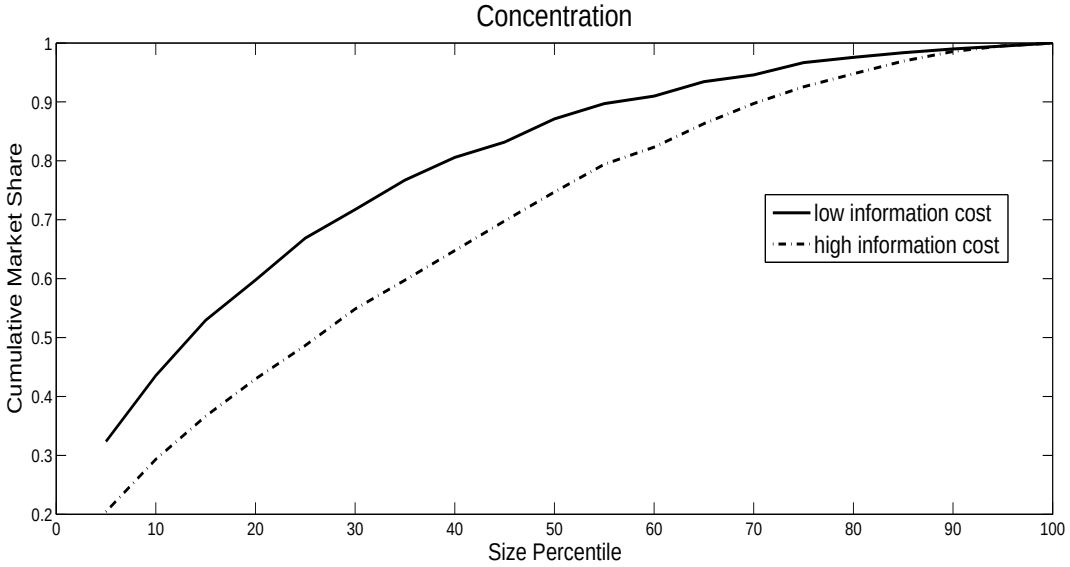


Figure 7: Scale of information cost: Distribution of sales across firm sizes ("0"=largest)

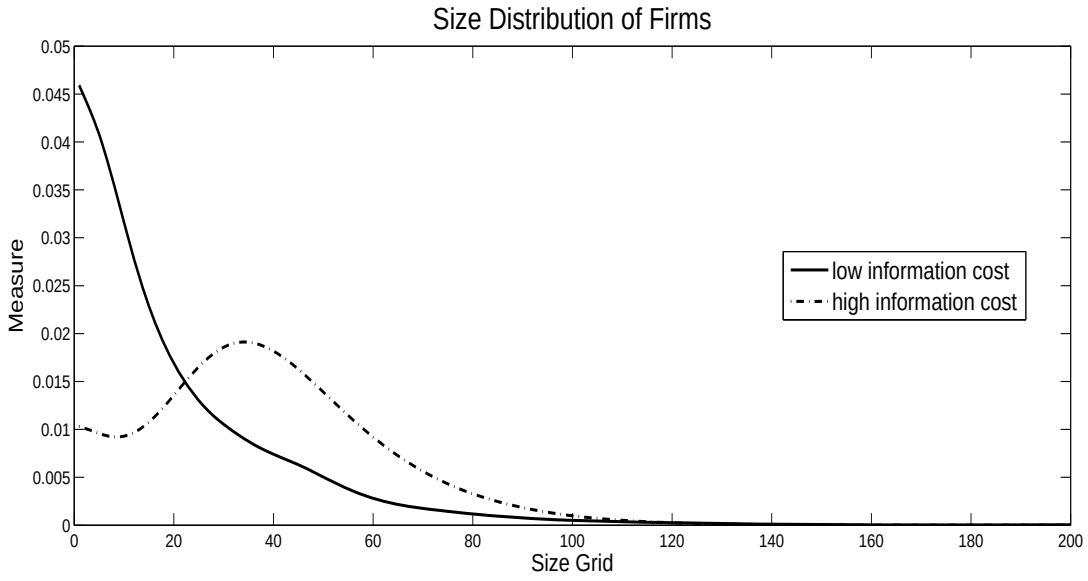


Figure 8: Scale of information cost: Density of firm size ("0"=smallest)

	Dependent Variable: 1997 Advertising Expenditures ($K)			
	I		II	
Independent variables:	OLS	Tobit	OLS	Tobit
1992 Value of Shipments ($K)	0.017***	0.015***	-	-
	[0.004]	[0.002]		
1992 Number of Employees	-	-	4.775***	5.127***
			[0.801]	[0.047]
1997 TFP	0.237***	2.114**	0.112**	1.988***
	[0.071]	[0.603]	[0.045]	[0.497]
Multi-unit dummy	278.5**	18,313.7***	345.1**	18,114.2***
	[122.0]	[133.5]	[127.6]	[145.2]
Industry fixed effects	Y	Y	Y	Y
Inverse Mills Ratio (IMR)	0.168***	-	0.177***	-
	[0.033]		[0.032]	
R^2 / *Log Likelihood*	0.33	-207,891	0.27	-208,337

Notes: All variables are in levels. OLS estimation is based on Heckman's two-step correction. IMR is the ratio of the standard normal density to the standard normal c.d.f. predicted from the selection equation.
Coefficients in OLS estimation (except for that of IMR) are marginal effects that take selection into account.
Corrected standard errors in parentheses. (*), (**), and (***) indicate significance at 10%, 5%, and 1% respectively.
Industry fixed effects are based on 1997 SIC codes (4-digit).

Table 1. The relationship between firm advertising expenditures, size, and productivity (CM Sample)

	Dependent Variable: 1992 Price ($K/unit)					
Independent variables:	I	II	III	IV	V	VI
1987 Value of Shipments ($K)	0.015**	-	-	0.063***	-	-
	[0.007]			[0.011]		
1987 Physical Quantity	-	0.012**	-	-	0.020**	-
		[0.005]			[0.008]	
1987 Employment	-	-	0.022***	-	-	0.070***
			[0.008]			[0.011]
1992 TFP (Quantity based)	-0.537***	-0.526***	-0.536***	-	-	-
	[0.050]	[0.051]	[0.050]			
1992 Average Variable Cost ($K)	-	-	-	0.140***	0.112***	0.134***
				[0.018]	[0.018]	[0.017]
Multi-unit dummy	0.122***	0.002	0.141***	-0.077***	0.136***	-0.074***
	[0.030]	[0.017]	[0.031]	[0.021]	[0.032]	[0.020]
Product fixed effects	Y	Y	Y	Y	Y	Y
R^2	0.96	0.95	0.96	0.95	0.95	0.94

Notes: All variables are in logarithms except for dummies. Robust standard errors in parantheses.
(*), (**), and (***) indicate significance at 10%, 5%, and 1%, respectively.

Table 2. The relationship between firm price, size, and productivity

	HHI (U.S.)			CR4 (U.S.)		
	Average HHI (MSA)[1]			Average CR4 (MSA)[1]		
Industry	**1992**	**1997**	**2002**	**1992**	**1997**	**2002**
New Car Dealers	2	2	15	1.5	1.5	5.6
	829	874	936	42.7	44.3	46.4
	[33.9]	[36.8]	[41.0]	[1.2]	[1.3]	[1.2]
Book Stores	828	1,151	1,533	41.3	54.1	65.6
	2,392	2,768	3,187	77.6	81.8	86.6
	[149.5]	[91.1]	[93.6]	[1.0]	[0.8]	[0.6]
Camera & Photo Supplies Stores	215	497	1,270	25.4	32.7	54.3
	6,016	6,040	6,467	94.3	95.9	97.4
	[190.7]	[178.8]	[172.3]	[0.8]	[0.6]	[0.4]
Tape, CD & Record Stores	559	829	1,000	40.9	46.8	57.7
	3,384	3,555	4,315	88.2	88.7	92.2
	[110.2]	[112.6]	[134.3]	[0.8]	[0.8]	[0.7]
Travel Agencies	160	115	574	12.7	16.3	20.9
	2,102	1,639	2,515	63.2	61.7	73.5
	[83.3]	[76.7]	[120.7]	[1.5]	[1.4]	[1.3]
Life Insurance	282	238	288	24.5	21.0	24.0
	3,422	3,810	4,836	79.7	84.1	89.0
	[154.4]	[154.7]	[181.4]	[1.2]	[1.1]	[0.9]

Notes: (1) Standard error of the average concentration measure across MSAs is in brackets. The label "U.S." means the statistics are nationwide. The label "U.S." indicates the statistics are nationwide. The label "MSA" indicates the statistics are calculated across MSAs only.

Table 3. The evolution of industry concentration for selected industries

	Skewness (U.S)			Average. Volatility[1] (U.S.)		No. of Firms (U.S)		
Industry	1992	1997	2002	1995-2000	2001-2007	1992	1997	2002
New Car Dealers	32.1	23.2	95.6	4,958.3 [29.4]	5,548.6 [20.7]	23,110	23,680	23,811
Book Stores	43.3	58.3	75.5	139.3 [1.5]	160.1 [1.7]	7,714	6,716	3,326
Camera & Photo Supplies Stores	23.7	29.9	45.2	202.5 [3.9]	276.1 [12.5]	1,931	1,502	1,254
Tape, CD & Record Stores	28.7	36.8	45.3	168.3 [5.2]	197.8 [6.4]	3,300	3,456	3,293
Travel Agencies	115.0	105.9	122.1	471.5 [5.1]	527.4 [2.1]	21,939	29,332	21,705
Life Insurance	18.8	13.2	12.4	171.6 [2.2]	194.2 [5.1]	1,035	1,012	937

Notes: (1) Standard error of the volatility measure over years is in brackets.
The label "U.S." indicates the statistics are nationwide.

Table 4. The evolution of industry aggregates for selected industries

Information and Industry Dynamics

Emin M. Dinlersoz and Mehmet Yorukoglu

Web Appendix

Abstract

This Web Appendix contains additional material accompanying the article "Information and Industry Dynamics" by Emin M. Dinlersoz and Mehmet Yorukoglu. Equation numbers in this Web Appendix continue the numbering in the article.

A Proofs

For notational convenience, define $R^* \equiv R^*(m) = \beta \frac{\partial}{\partial m} \int V^*(m,c)h(c)dc$ as the derivative of the continuation value of a firm with respect to its size m at the beginning of the next period. Given $A \subseteq \mathbb{R}^n$ $(n \geq 1)$, $B \subseteq \mathbb{R}$, and any two functions $f, g : A \to B$, $f \preceq g$ means $f(\mathbf{x}) \leq g(\mathbf{x})$ for all $\mathbf{x} \in A$. Similarly, $f \prec g$ means $f(\mathbf{x}) \leq g(\mathbf{x})$ for all $\mathbf{x} \in A$ and $f(\mathbf{x}) < g(\mathbf{x})$ for some $\mathbf{x} \in A$. Whenever there is no ambiguity, the arguments of a function are suppressed.

Proof of Lemma 1. z_t is defined by (4), and the demand ω_t of a consumer who did not purchase from firm (p, m^o) in period $t-1$ is

$$\omega_t(p, m^o) = \sum_{i=0}^{\infty} \Omega_t(i) \widetilde{G}_t^i(W_t(p, m^o)), \tag{19}$$

where $\Omega_t(i)$ is the measure of consumers informed of $i \geq 0$ firms in addition to firm (p, m^o), and $\widetilde{G}_t^i$ is the probability that the consumer chooses to buy from firm (p, m^o). Let $\delta_t < 1$ denote the measure of consumers who purchased in period $t-1$ from a firm that exited at the beginning of period t. Let $\theta_t = \lambda + (1-\lambda)\delta_t < 1$ be the sum of the mass of new consumers and the mass of surviving consumers whose firms exited at the beginning of period t. Label all of these consumers as 'unattached', since they have no information before they receive any ads. Label the rest of the consumers as 'attached', as they are informed of their most-recently-visited firms before they receive any ads. Then,

$$\Omega_t(i) = \begin{cases} \theta_t \Psi_t(0) & \text{for } i = 0, \\ \theta_t \Psi_t(i) + (1-\theta_t)\Psi_t(i-1) & \text{for } i > 0. \end{cases} \tag{20}$$

In (20), the term $\theta_t \Psi_t(0)$ gives the mass of unattached consumers who receive zero ads. The term for $i > 0$ is the mass of unattached consumers who receive i ads, plus the mass of attached consumers who receive $(i-1)$ ads. Next, let $\alpha_t \equiv \alpha_t(p, m^o)$ be the probability that a consumer prefers firm (p, m^o) to the firm he purchased from in period $t-1$. Note that $G_t^0 = \widetilde{G}_t^0 = 1$. For $i \geq 1$, (20) implies

$$\widetilde{G}_t^i = \frac{\theta_t \Psi_t(i)}{\Omega_t(i)} G_t^i + \frac{(1-\theta_t)\Psi_t(i-1)}{\Omega_t(i)} G_t^{i-1} \alpha_t, \tag{21}$$

where the first term in (21) is the probability that a consumer is unattached and has i new ads conditional on having information about i other firms, and the second term is the probability

that a consumer is attached and has $i-1$ new ads conditional on having information about i other firms. Using (19) and (21), one can then write

$$\omega_t(p, m^o) = \theta_t \sum_{i=0}^{\infty} \Psi_t(i) G_t^i + (1-\theta_t) \sum_{i=1}^{\infty} \Psi_t(i-1) G_t^{i-1} \alpha_t = [\theta_t + (1-\theta_t)\alpha_t] z_t(p, m^o). \quad (22)$$

Because $\alpha_t \leq 1$ and $\theta_t < 1$, it follows that $\omega_t(p, m^o) \leq z_t(p, m^o)$. Next, for $\alpha_t(p, m^o) < 1$, using (22) and differentiating with respect to price and rearranging yields

$$\varepsilon_{\omega_t}(p, m^o) = \frac{(1-\theta_t)\alpha_t(p, m^o)}{\theta_t + (1-\theta_t)\alpha_t(p, m^o)} \varepsilon_{\alpha_t}(p, m^o) + \varepsilon_{z_t}(p, m^o),$$

which implies $\varepsilon_{\omega_t}(p, m^o) > \varepsilon_{z_t}(p, m^o)$ because $\varepsilon_{\alpha_t}(p, m^o) > 0$ and $(1-\theta_t)\alpha_t(p, m^o) > 0$. For $\alpha_t(p, m^o) = 1$, (22) implies $\varepsilon_{\omega_t}(p, m^o) = \varepsilon_{z_t}(p, m^o)$. Consequently, $\varepsilon_{\omega_t}(p, m^o) \geq \varepsilon_{z_t}(p, m^o)$. ■

Proof of Proposition 1. Part (i). Let $Q^*(p|m^o)$ denote the equilibrium probability that an ad contains a price of at least p, conditional on m^o. The cumulative distribution function, $1 - Q^*(p|m^o)$, cannot have a mass at any p in its support, otherwise any firm charging p could reduce its price slightly and steal a positive mass of consumers from other firms charging p, leading to a discrete gain in firm value. Also, $1 - Q^*(p|m^o)$ cannot be flat over some interval (p_1, p_2), otherwise any firm charging p_1 could increase its price to p_2 without a reduction in its probability of sale. Thus, $1 - Q^*(p|m^o)$ is strictly increasing in p over the interior of its support $[\underline{p}^*(m^o), \overline{p}^*(m^o)]$, where $\underline{p}^*(m^o)$ and $\overline{p}^*(m^o)$ are the minimum and maximum prices observed for firms with size m^o. Therefore, $\frac{\partial Q^*(p|m^o)}{\partial p} < 0$. Let $z^*(p|m^o) = \sum_{i=0}^{\infty} \Psi^*(i) \left[Q^*(p|m^o)\right]^i$ be the demand function for a consumer conditional on m^o. It follows that $\frac{\partial z^*(p|m^o)}{\partial p} = \sum_{i=0}^{\infty} \Psi^*(i) i Q^{*i-1} \frac{\partial Q^*}{\partial p} < 0$, which also implies $\frac{\partial z^*(p, m^o)}{\partial p} < 0$. Similar arguments apply to $\omega^*(p, m^o)$.

Parts (ii), (iii) and (iv). By the envelope theorem, $\frac{dV^*}{dc} = \frac{\partial V^*}{\partial c} = -m^* < 0$. Next, observe that

$$\frac{\partial z^*}{\partial p} = \left(\sum_{i=0}^{\infty} \Psi^*(i) i G^{*i-1} \right) G^{*\prime} \frac{\partial W^*}{\partial p}. \quad (23)$$

Because $\frac{\partial z^*}{\partial p} < 0$ by Part (i) and $\left(\sum_{i=0}^{\infty} \Psi^*(i) i G^{*i-1}\right) G^{*\prime} > 0$, it follows that $\frac{\partial W^*}{\partial p} < 0$. Next, we argue that z^*, ω^*, V^*, and W^* must be strictly increasing in m^o. Differentiation of z^* yields

$$\frac{\partial z^*}{\partial m^o} = \left(\sum_{i=0}^{\infty} \Psi^*(i) i G^{*i-1} \right) G^{*\prime} \frac{\partial W^*}{\partial m^o}. \quad (24)$$

Thus, $\frac{\partial z^*}{\partial m^o}$ and $\frac{\partial W^*}{\partial m^o}$ have the same sign. Similarly, $\frac{\partial \omega^*}{\partial m^o}$ and $\frac{\partial W^*}{\partial m^o}$ have the same sign, and therefore so do $\frac{\partial \omega^*}{\partial m^o}$ and $\frac{\partial z^*}{\partial m^o}$. Furthermore, the envelope theorem implies

$$\frac{dV^*}{dm^o} = \frac{\partial V^*}{\partial m^o} = \frac{\partial m^*}{\partial m^o}\left(p^* - c + R^*\right). \tag{25}$$

For (11) to hold, the term in parentheses in (25) must be positive. Consequently, $\frac{dV^*}{dm^o}$ and $\frac{\partial m^*}{\partial m^o}$ have the same sign. Next, note that

$$\frac{\partial m^*}{\partial m^o} = (1-\lambda)\, z^* + \left(m^o(1-\lambda)\frac{\partial z^*}{\partial m^o} + a^*\frac{\partial \omega^*}{\partial m^o}\right).$$

To obtain a contradiction, suppose that $\frac{\partial z^*}{\partial m^o} \leq 0$ and $\frac{\partial \omega^*}{\partial m^o} \leq 0$. There are then two possibilities: $\frac{\partial m^*}{\partial m^o} \leq 0$ or $\frac{\partial m^*}{\partial m^o} > 0$. If $\frac{\partial m^*}{\partial m^o} \leq 0$, (25) implies $\frac{dV^*}{dm^o} \leq 0$. But then firm value is non-increasing in firm size, and a firm has no incentive to increase its size, because exit probability increases as firm size increases. Because the only way a firm can grow is to send ads, there is then no advertising. Therefore, $\frac{\partial z^*}{\partial m^o} \leq 0$, $\frac{\partial \omega^*}{\partial m^o} \leq 0$, and $\frac{\partial m^*}{\partial m^o} \leq 0$ cannot hold in an equilibrium with positive advertising.

Suppose now that $\frac{\partial m^*}{\partial m^o} > 0$. Then, by (25), $\frac{dV^*}{dm^o} > 0$. Now, rewrite the consumer surplus in (1) in stationary equilibrium as

$$W^*(p, m^o) = U(p) + \beta(1-\lambda)\sum_{i=0}^{\infty}\Psi^*(i)\{(1 - X^*(p, m^o))W_1^*(p^1, m^1) + X^*(p, m^o)W_2^*]\},$$

where X^* is the probability of exit, (p^1, m^1) denotes the next period price and size, and

$$W_1^*(p^1, m^1) = \begin{cases} E\left[\int \max\{W, W^*(p^1, m^1)\} dG^{*i}(W)\right] & \text{if } i > 0 \\ E[W(p', m')] & \text{if } i = 0 \end{cases},$$

$$W_2^* = \begin{cases} E\left[\int W dG^{*i}(W)\right] & \text{if } i > 0 \\ W^{*\varnothing} & \text{if } i = 0 \end{cases}.$$

Then

$$\frac{\partial W^*}{\partial m^o} = \beta(1-\lambda)\sum_{i=0}^{\infty}\Psi^*(i)\left(\frac{\partial X^*}{\partial m^o}(W_2^* - W_1^*(p^1, m^1)) + (1 - X^*)\frac{\partial W_1^*}{\partial m^1}\right), \tag{26}$$

and

$$\frac{\partial W_1^*}{\partial m^1} = \beta(1-\lambda)\sum_{i=0}^{\infty}\Psi^*(i)\left(\frac{\partial X^*}{\partial m^1}(W_2^* - W_1^*(p^2, m^2)) + (1 - X^*)\frac{\partial W_1^*}{\partial m^2}\right)\frac{\partial m^1}{\partial m^o}, \tag{27}$$

where (p^2, m^2) is the two-period ahead price and size, and m^1 is the next period's size. One can thus substitute $\frac{\partial W_1^*}{\partial m^1}$ in (26) using (27). Continuing with similar substitution for all future periods, one obtains

$$\frac{\partial W^*}{\partial m^o} = \sum_{t=1}^{\infty} \beta^t (1-\lambda)^t \left(\prod_{j=1}^{t-1} (1 - X^*(p^t, m^t)) \right) \frac{\partial X^*}{\partial m^t} \sum_{i=0}^{\infty} \Psi^*(i)(W_2^* - W_1^*(p^t, m^t)) \left(\prod_{j=1}^{t-1} \frac{\partial m^j}{\partial m^{j-1}} \right), \tag{28}$$

where (p^t, m^t) is the t-period ahead price and size, and m^j is the size in period j. Because $\frac{dV^*}{dm^o} > 0$ and exit probability declines with firm value, exit probability also declines with firm size, i.e. $\frac{\partial X^*}{\partial m^t} < 0$ for all t. In addition, it was assumed that $\frac{\partial m^j}{\partial m^{j-1}} > 0$. Furthermore, $W_2^* - W_1^*(p^1, m^1) < 0$, by the definitions of W_1^* and W_2^*. Consequently, (28) implies $\frac{\partial W^*}{\partial m^o} > 0$. But since $\frac{\partial z^*}{\partial m^o} \leq 0$, (24) implies $\frac{\partial W^*}{\partial m^o} \leq 0$, a contradiction. Therefore, $\frac{\partial z^*}{\partial m^o} \leq 0$ and $\frac{\partial \omega^*}{\partial m^o} \leq 0$ cannot hold. Thus, the only configuration consistent with positive advertising in equilibrium is $\frac{\partial z^*}{\partial m^o} > 0$, $\frac{\partial \omega^*}{\partial m^o} > 0$, $\frac{\partial V^*}{\partial m^o} > 0$ and $\frac{\partial W^*}{\partial m^o} > 0$. ■

Proof of Proposition 2. Part (i). Proposition 1(i) established that equilibrium entails a continuous (atomless) price distribution. The firm type that offers the least surplus to consumers cannot steal any consumers away from other firms and can do no better than charging its monopoly price. For other firm types, optimal price must satisfy (11). To see this claim, note that, by Proposition 1, the individual demand functions, $z^*(p, m^o)$ and $\omega^*(p, m^o)$, and hence a firm's residual demand $m^*(p, a; m^o)$, are continuous in p, implying the continuity of the profit function and the value function in p for surviving firms. Furthermore, feasible prices for a firm type can be restricted to a compact set $[\underline{p}, p^m]$, where $p^m \geq s$ is the monopoly price for the corresponding firm type, and a lower bound $\underline{p}$ can be imposed because, as price declines, $p - c + R^*$ becomes negative eventually as R^* is bounded from above. Therefore, for firms that offer more than the lowest surplus, a value maximizing price lies in the set $(\underline{p}, p^m)$. Given the continuity of m^* in p, lowering price entails a trade-off: a decline in profit per consumer (including a firm's captive consumers who are only informed of a single firm) versus an increase in profits due to a higher probability of sale to consumers who have ads from other firms. The first order condition (11) balances these two trade-offs. To see uniqueness, suppose that the value function for firm (m^o, c) admits two or more countable interior global maximizers. Take any two such maximizers, (p_1, a_1) and

(p_2, a_2), such that $p_1 \neq p_2$. Since (11) holds at both (p_1, a_1) and (p_2, a_2), by the generalized intermediate value theorem (see, e.g. Munkres (1975) p. 154) there must exist a pair (p', a') $\neq (p_1, a_1)$, (p_2, a_2) such that

$$[m^o(1-\lambda)z^*(p', m^o)+a'\omega^*(p', m^o)]+\left[m^o(1-\lambda)\frac{\partial z^*(p', m^o)}{\partial p}+a'\frac{\partial \omega^*(p', m^o)}{\partial p}\right](p'-c+R^*)=0. \tag{29}$$

Because the distribution of price and advertising pairs is continuous conditional on m^o, there must exist some firm (m^o, c') with $c' \neq c$ for which (p', a') is optimal. For this firm, (p', a') satisfies

$$[m^o(1-\lambda)z^*(p', m^o)+a'\omega^*(p', m^o)]+\left[m^o(1-\lambda)\frac{\partial z^*(p', m^o)}{\partial p}+a'\frac{\partial \omega^*(p', m^o)}{\partial p}\right](p'-c'+R^*)=0. \tag{30}$$

But because $c \neq c'$ and all other terms are identical in (29) and (30), the two equalities cannot hold simultaneously. Thus, there cannot be more than one discrete global interior maximizer. The remaining possibility is that there exists a continuum of maximizers for firm (m^o, c) that form a connected set S of (p, a) pairs, and no other firm type has a value-maximizing price and advertising pair within this set. Suppose that is the case. Now consider any two value-maximizing pairs (p_1, a_1) and (p_2, a_2) in this set such that $p_1 < p_2$. Because it is assumed that there is no other firm type for which some pair $(p, a) \in S$ is a value maximizer, firm (m^o, c) can raise its price from p_1 to p_2 and still send a_1 ads without a reduction in its residual demand, and thereby increase its value. This contradicts with (p_1, a_1) being part of the set S.

Taking the total derivative of (11) and (13) with respect to c leads to

$$\frac{\partial p^*}{\partial c}=-\frac{1}{\Delta}\left[\Phi''\left(m^o(1-\lambda)\frac{\partial z^*}{\partial p}+a^*\frac{\partial \omega^*}{\partial p}\right)+\left(\omega^*+\frac{\partial \omega^*}{\partial p}(p^*-c+R^*)\right)\right], \tag{31}$$

where

$$\begin{aligned}\Delta \;=\; & \left(\omega^*\tfrac{dR^*}{da}-\Phi''\right)\left[\left(m^o(1-\lambda)\tfrac{\partial z^*}{\partial p}+a^*\tfrac{\partial \omega^*}{\partial p}\right)\left(2+\tfrac{\partial R^*}{\partial p}\right)+\left(m^o(1-\lambda)\tfrac{\partial^2 z^*}{\partial p^2}+a\tfrac{\partial^2 \omega^*}{\partial p^2}\right)(p^*-c+R^*)\right] \\ & -\left(\tfrac{\partial \omega^*}{\partial p}(p^*-c+R^*)+\omega^*(1+\tfrac{\partial R^*}{\partial p})\right)^2\end{aligned}$$

is the determinant of the Hessian of V^*. Note that $\Delta > 0$, given the fact that the Hessian of V^* is negative definite at the global maximizer (p^*, a^*). Inside the square brackets in (31), the

first term is negative because $\Phi'' > 0$ (Assumption 1), $\frac{\partial z^*}{\partial p} < 0$ and $\frac{\partial \omega^*}{\partial p} < 0$ (Proposition 1). To sign the second term inside the square brackets in (31), note that (5) and (11) together yield

$$\omega^* + \frac{\partial \omega^*}{\partial p}(p^* - c + R^*) = \frac{(1-\theta^*)\frac{\partial \alpha^*}{\partial p} z^*(p^* - c + R^*)}{1 + [\theta^* + (1-\theta^*)\alpha^*]\frac{a^*}{m^o(1-\lambda)}} < 0, \tag{32}$$

because $\frac{\partial \alpha^*}{\partial p} < 0$ and all other terms in the ratio on the r.h.s. of (32) are positive. Consequently, the expression in square brackets in (31) is negative, implying $\frac{\partial p^*}{\partial c} > 0$.

To sign $\frac{\partial p^*}{\partial m^o}$, suppose, to obtain a contradiction, that for two firm types with marginal cost c and sizes $m_2^o > m_1^o$, it holds that $p^*(m_2^o, c) \leq p^*(m_1^o, c)$. Given the continuity of the pricing policy, one can find a firm type (m_3^o, c) where $m_3^o \in [m_1^o, m_2^o]$ such that $p^*(m_3^o, c) = p^*(m_2^o, c)$ and $p^*(m^o, c)$ is non-increasing over the interval $(m_3^o - \varepsilon, m_3^o)$ for some $\varepsilon > 0$. Then, by Proposition 1(iv), $W^*(p^*(m_3^o, c), m_3^o) > W^*(p^*(m^o, c), m^o)$ for all $m^o \in [m_3^o - \varepsilon, m_3^o)$. Therefore, firm (m_3^o, c) can raise its price slightly without a reduction in its probability of sale and increase its profit per consumer, a contradiction with $p^*(m_3^o, c)$ being a value maximizer for firm (m_3^o, c).

Part (ii). If the exit probability $X^*(p, m^o)$ does not depend on m^o directly, $W^*(p, m^o)$ also does not depend on m^o directly, and it reduces to a function, $W^*(p)$, of price only. At this point, we allow for the possibility that in equilibrium $W^*(p^*)$ depends on firm size indirectly through the optimal price p^*. By definitions (4) and (5), the demand functions $z^*(p, m^o)$ and $\omega^*(p, m^o)$ are then also functions of price only: $z^*(p)$ and $\omega^*(p)$. Because a larger size now confers no additional surplus to any consumer, the demand functions for previous-period customers and the new customers acquired through ads must also be identical, i.e. $z^*(p) \equiv \omega^*(p)$. Consequently, (11) reduces to

$$(m^o(1-\lambda) + a)\, z^* + (m^o(1-\lambda) + a)\frac{dz^*}{dp}\,(p - c + R^*) = 0.$$

Dividing through by $m^o(1-\lambda) + a > 0$ and rearranging terms, a firm's optimal price can be written as

$$p^* = \frac{\varepsilon_{z^*}(p^*)}{\varepsilon_{z^*}(p^*) - 1}(c - R^*).$$

The price depends on the elasticity $\varepsilon_{z^*}(p^*)$ of the individual demand function $z^*(p^*)$, which is independent of firm size. Furthermore, R^*, which is the change in the continuation value of

the firm due to a change in its size m^* at the beginning of the next period, is also independent of m^o. Consequently, price is not a function of m^o. ■

Proof of Proposition 3. The proof of Proposition 2(i) and the strict convexity of Φ (Assumption 1(i)) imply that a unique value-maximizing $a^*(m^o, c)$ exists. Taking the total derivative of (11) and (13) with respect to c yields

$$\frac{\partial a^*}{\partial c} = -\frac{1}{\Delta}\left[\left(m^o(1-\lambda)\frac{\partial z^*}{\partial p} + a^*\frac{\partial \omega^*}{\partial p}\right)\left(\omega^* + \frac{\partial \omega^*}{\partial p}(p^* - c + R^*)\right) - \omega^* Y^*\right], \tag{33}$$

where

$$Y^* = 2\left(m^o(1-\lambda)\frac{\partial z^*}{\partial p} + a^*\frac{\partial \omega^*}{\partial p}\right) + \left(m^o(1-\lambda)\frac{\partial^2 z}{\partial p^2} + a^*\frac{\partial^2 \omega^*}{\partial p^2}\right)(p^* - c + R^*).$$

To sign (33), note that $\Delta > 0$, as shown earlier. Second, the first term inside the brackets in (33) is positive, following from (32) and that $\frac{\partial z^*}{\partial p} < 0$ and $\frac{\partial \omega^*}{\partial p} < 0$ (Proposition 1). Next, $\Delta > 0$ and the properties of the Hessian of V^* at (p^*, a^*) imply

$$Y^* < \frac{\left(\frac{\partial \omega^*}{\partial p}(p^* - c + R^*) + \omega^*(1 + \frac{\partial R^*}{\partial p})\right)^2}{\omega^* \frac{\partial R^*}{\partial a} - \Phi''} - \frac{1}{\Phi''}\omega^{*2}\left(\frac{\partial R^*}{\partial p}\right)^2 < 0,$$

where the sign follows because $\omega^* \frac{\partial R^*}{\partial a} - \Phi'' < 0$ (because $\Delta > 0$) and $\Phi'' > 0$ (Assumption 1). As a result, the second term inside the brackets in (33) is negative. Consequently, the entire term inside the brackets in (33) is positive. Thus, (33) is negative.

To see the monotonicity of a^* in m^o, consider two firms (m_1^o, c) and (m_2^o, c) such that $m_1^o < m_2^o$. Firm (m_2^o, c) has a higher expected return in the current period from a marginal ad

$$\omega^*(p_1^*, m_1^o)(p_1^* - c) < \omega^*(p_2^*, m_2^o)(p_2^* - c).$$

To see the last inequality, note that $p_2^* > p_1^*$ (Proposition 2). Furthermore, it must be that $\omega^*(p_1^*, m_1^o) < \omega^*(p_2^*, m_2^o)$. Suppose not. Firm (m_2^o, c) could then charge price p_1^* and have a probability of sale per ad $\omega^*(p_1^*, m_2^o) > \omega^*(p_1^*, m_1^o)$ by Proposition 1. Firm (m_2^o, c) could then afford to raise its price and still have a higher probability of sale and higher profit per ad. Thus, the marginal ad brings higher marginal profit in the current period to firm (m_2^o, c). Because firm (m_2^o, c) also has a lower likelihood of exit for all future periods, the consumer acquired through the marginal ad also brings a larger expected marginal benefit to firm

(m_2^o, c) from the next period onwards. The value effect of the marginal ad must therefore be larger for firm (m_2^o, c). Then, by (13) and the fact that Φ' is strictly increasing, firm (m_2^o, c) must send more ads. ■

Proof of Proposition 4. The properties of m^* follow directly from its definition and the properties of a^* and z^*. Since both m^* and p^* are strictly increasing in m^o, so is r^*. To see the monotonicity of r^* in c, note that

$$\frac{\partial r^*}{\partial c} = \frac{\partial a^*}{\partial c}\omega^* p^* + \left[\left((1-\lambda)m^o\frac{\partial z^*}{\partial p} + a^*\frac{\partial \omega^*}{\partial p}\right)p^* + ((1-\lambda)m^o z^* + a^*\omega^*)\right]\frac{\partial p^*}{\partial c} \tag{34}$$

Rearranging (11), and then substituting it in (34), we obtain

$$\begin{aligned}\frac{\partial r^*}{\partial c} &= \frac{\partial a^*}{\partial c}\omega^* p^* + \left((1-\lambda)m^o\frac{\partial z^*}{\partial p} + a^*\frac{\partial \omega^*}{\partial p}\right)(c - R^*)\frac{\partial p^*}{\partial c} \\ &< \frac{\partial a^*}{\partial c}\omega^* p^* + \left((1-\lambda)m^o\frac{\partial z^*}{\partial p} + a^*\frac{\partial \omega^*}{\partial p}\right)p^*\frac{\partial p^*}{\partial c} < 0,\end{aligned}$$

where the first inequality follows because $p^* - c + R^* > 0$, and the second inequality from the fact that $\frac{\partial a^*}{\partial c} < 0$, $\frac{\partial z^*}{\partial p} < 0$, $\frac{\partial \omega^*}{\partial p} < 0$, and $\frac{\partial p^*}{\partial c} > 0$. ■

Theorem 1. Given the values of the parameters of the model except κ, there is some $\kappa' > 0$ such that there exists a unique stationary equilibrium with a positive advertising and positive entry and exit, i.e. $M^* > 0$ and $x^*(c) > 0$ for some $c \in [\underline{c}, \infty)$, as long as $\kappa < \kappa'$.

Proof of Theorem 1. Given any equilibrium measure of firms μ^*, the exit threshold $c^*(m^o)$ is uniquely defined for any $m^o \geq 0$ by Proposition 1. Since $V^*(m^o, c) = 0$ for $c \geq c^*(m^o)$, we can write for any m^o

$$\int_{c^*(m^o)}^{\infty} V^*(m^o, c)h(c)dc = 0. \tag{35}$$

For positive entry in equilibrium, free entry condition (9) must hold with equality

$$\int_{\underline{c}}^{\infty} V^*(0, c)h(c)dc = \kappa. \tag{36}$$

Let $S = \{(m', c') : m' = m^*(m^o, c),\ c' \leq c^*(m')\}$ be the set of firm of types reachable from firm type (m^o, c), conditional on staying in the industry. Define the operator L_{c^*} as

$$L_{c^*}(m^o, c; S) = \begin{cases} \int_S h(c')dc', \text{if } c \leq c^*(m^o) \\ \qquad 0, \text{ otherwise,} \end{cases}$$

with norm $\|L_{c^*}\| < 1$. Invariance of μ^* in stationary equilibrium requires

$$\mu^* = L_{c^*}\mu^* + M^*H, \tag{37}$$

where $L_{c^*}\mu^*(S) = \int L_{c^*}(m^o, c; S)d\mu^*$. A stationary equilibrium with positive entry and exit is then given by a triplet $(c^*(\cdot), M^*, \mu^*)$ that satisfies equations (35)-(37) simultaneously.

Next, consider the advertising equilibrium corresponding to a given measure μ, an entry mass M, and an exit rule $c(\cdot)$. Let $E = \{\mu, M, c(\cdot)\}$. Let A denote the space of continuous, bounded functions defined over firm types under E, i.e. $a(E) \equiv a(\cdot, \cdot; E) : [\underline{m}, \overline{m}] \times [\underline{c}, \infty) \to \mathbb{R}^+$, where $\underline{m} > 0$ and $\overline{m} < \infty$ are the minimum and maximum firm sizes under E. Endow A with the sup-norm. Define $T : A \to A$ as the operator that matches any $a \in A$ to some $T(a) \in A$ that results from the firms' optimal choices of advertising given the distribution of consumer surplus $G(W)$ generated by the function a under E. We will first show that, corresponding to any E, there exists a unique advertising equilibrium such that the advertising policy $a(E)$ adopted by firms under E generates a distribution of consumer surplus across ads that renders the same advertising policy $a(E)$ optimal for E. We will then show the existence and uniqueness of a triplet $E^* = \{\mu^*, M^*, c^*(\cdot)\}$ that satisfies (35)-(37). The following lemmas accomplish these tasks.

Lemma 2. Given any E, T has a unique fixed point $a(E)$, i.e. $T(a(E)) = a(E)$.

Lemma 3. For any configuration of the model's parameters (except κ) that satisfies the model's assumptions, there exists some $\kappa' > 0$ such that, given any $\kappa < \kappa'$, there is a unique $E^* = \{\mu^*, M^*, c^*(\cdot)\}$ which satisfies (35)-(37).

Proof of Lemma 2. Given E and any $a \in A$, consider the cumulative advertising $A(m^o, c; E)$ made by firms that offer at most as much surplus as firm (m^o, c). In other words,

$$A(m^o, c; E) = \int_{\underline{m}}^{\overline{m}} \int_{c(x)}^{\infty} \Gamma(x, y, A(x, y; E))d\mu(x, y), \tag{38}$$

where $c(x)$ is the marginal cost level such firm $(x, c(x))$ offers as much surplus as firm (m^o, c) does. Note that $A(m^o, c; E)$ is continuous and monotonic in its arguments m^o and c. Using

(13), the integrand Γ is defined as

$$\begin{aligned}\Gamma(x,y,A(x,y;E)) &= a(x,y;E) \\ &= \Phi'^{-1}\left(\omega(p(x,y),x;A)\left[(p(x,y;A)-y)+\beta\int\frac{\partial V(m,c';A)}{\partial m}h(c')dc'\right]\right) \\ &= \Phi'^{-1}(\Lambda(x,y,A(x,y;E)).\end{aligned}$$

The dependence of the functions ω, p, and V on $A(x,y;E)$ is made explicit. Equation (38) is a Volterra-Fredholm type integral equation. We will show that (38) has a unique fixed point, $A^*(m^o,c;E)$, in the space of continuous, bounded, and non-decreasing functions defined over (m^o,c) pairs. For any $A_1(x,y;E)$ and $A_2(x,y;E)$ in that space, an application of Mean Value Theorem implies

$$\|\Gamma(x,y,A_1(x,y;E))-\Gamma(x,y,A_2(x,y;E))\| \leq \|D(x,y)\|\,\|A_1(x,y;E)-A_2(x,y;E)\|,$$

where $\|\cdot\|$ denotes the sup-norm, and

$$D(x,y)=\frac{1}{\Phi''(\Lambda(x,y,A(x,y))}\frac{\partial\Lambda(x,y,A(x,y))}{\partial A(x,y)}.$$

Γ is bounded and continuous in its arguments, because Φ'^{-1} and Λ are both continuous. $\|D(x,y;A)\|$ is also bounded, continuous, and integrable. Moreover the function $A(x,y;E)$ is bounded because $a(x,y;E)<\infty$ for all (x,y). Therefore, the conditions for the existence and uniqueness of a solution to the integral equation (38) are satisfied (See, e.g., Proposition 2 in Hacia (1997)). As a result, there exists a unique, continuous, non-decreasing $A(E)\equiv A(m^o,c;E)$ that satisfies (38). Consequently, the advertising function $a(E)\equiv a(m^o,c;E)$ associated with $A(E)$ also exists and it is unique. In other words, the operator $T(a)$ has a unique fixed point $a(E)$.

Proof of Lemma 3. Given the existence and uniqueness of an advertising equilibrium corresponding to a given E, we now focus on the existence and uniqueness of a triplet E satisfying equations (35)-(37). We build on the basic arguments in Hopenhayn (1992). Assumptions A3, A4 and A5 in Hopenhayn (1992) apply with little modification: A3(a) is satisfied because H is continuous in current period's cost shock, and H is invariant to previous period's shock, implying continuity. A3(b) is replaced with an invariant *i.i.d.* process here, but the fact that current cost shock affects end-of-period firm size implies that firm's

future value depends on the current cost shock, allowing the exit threshold to be defined as $c^*(m^o)$, as described in the text. A4 is also satisfied because in any period the probability, $1-H(c)$, of observing a cost shock greater than c is positive for any c. Finally, A5 is satisfied because the distribution of entrants' cost shocks, H, is continuous.

Existence: Let D be the set of continuous, bounded, non-decreasing functions defined as $c(\cdot) : \mathbb{R}^+ \to [\underline{c}, \infty)$. First, we show that, for any $c(\cdot) \in D$ and $M > 0$, an invariant measure $\mu(c(\cdot), M)$ exists, i.e.

$$\mu(c(\cdot), M) = L_c \mu(c(\cdot), M) + MH.$$

Equivalently, μ satisfies

$$\mu(c(\cdot), M) = M(\mathcal{I} - L_c)^{-1} H,$$

where I is the identity operator and $(I - L_c)^{-1}$ is the inverse operator for $I - L_c$. Following steps in Lemma 4 of Hopenhayn (1992), let L_c^n be the composition of L_c n times with itself and $L_c^0 = I$. The assumption that $H(c) < 0$ for all $c \in [\underline{c}, \infty)$ implies that the norm $\|L_c^n\| < 1$, so $(I - L_c^n)^{-1} = \sum_{t=0}^{\infty} L_c^{nt}$ following from Kolmogorov and Fomin (1970, Chapter 6, Section 23, Theorem 4, p. 231). The existence of $(I - L_c)^{-1}$ then follows because $\|L_c^n\|$ is non-increasing in n. Thus, $\mu(c(\cdot), M)$ satisfies invariance.

Next, we will say that the invariant measure $\mu(c(\cdot), M)$ is continuous on D if for all $c(\cdot) \in D$ and for every sequence $c_n(\cdot) \to c(\cdot)$, we have $\mu(c_n(\cdot), M) \to \mu(c(\cdot), M)$. Continuity with respect to M is defined similarly, but simply on $\mathbb{R}^+$. Using arguments similar to Lemma 5 in Hopenhayn (1992), it can be shown that the invariant measure $\mu(c(\cdot), M)$ is jointly continuous, strictly increasing in M, and non-increasing in $c(\cdot)$, i.e. for two exit schedules $c_1(\cdot)$, $c_2(\cdot) \in D$ such that $c_2(m^o) \succ c_1(m^o)$, we have $\mu(c_2(\cdot), M) \leq \mu(c_1(\cdot), M)$ in the sense that $\int_c u(m^o, c) d\mu_2(m^o, \cdot) \leq \int_c u(m^o, c) d\mu_1(m^o, \cdot)$ for any non-decreasing function u and for any given m^o.

For any exit rule $c(\cdot)$, define the entry mass $M^e(c(\cdot))$ implicitly as

$$V^e(\mu(c(\cdot), M^e)) = \int_{\underline{c}}^{\infty} V^* \left(0, c; \mu(c(\cdot), M^e)\right) h(c) dc = \kappa.$$

In other words, for the invariant measure μ, $M^e(c(\cdot))$ is the mass of entrants that are needed for the expected discounted profit for entrants, V^e, to be equal to the cost of entry under

the exit rule $c(\cdot)$. Also, define $M^x(c(\cdot))$ as the mass of entrants such that for the invariant measure μ, the exit rule $c(\cdot)$ is optimal, i.e. for all $m^o \geq 0$

$$\int_{c(m^o)}^{\infty} V^*(m^o, c; \mu(c(\cdot), M^x))h(c)dc = 0.$$

A stationary equilibrium with positive entry and exit exists if and only if there is a function $c^*(\cdot)$ such that $M^e(c^*(\cdot)) = M^x(c^*(\cdot))$. This amounts to showing that the functions M^e and M^x intersect at least once.

It can be shown, analogous to Lemma 6 in Hopenhayn (1992), that the function $M^x : D \to \mathbb{R}^+$ is well-defined, continuous and strictly increasing, i.e. for any two functions $c_1(\cdot)$, $c_2(\cdot) \in D$ such that $c_2(\cdot) \succ c_1(\cdot)$, we have $M^x(c_2(\cdot)) > M^x(c_1(\cdot))$. In particular, let $Z(m^o, c; c(\cdot), M) = \int_{c(m^o)}^{\infty} V(m^o, c; \mu(c(\cdot), M))h(c)dc$. It can be shown that Z is strictly increasing in m^o, strictly decreasing in c, non-decreasing in $c(\cdot)$ and strictly decreasing in M. Note that as $M \to \infty$, $Z(m^o, c; c(\cdot), M) \to -\frac{1}{1-\beta}f$, because $\Pi(m^o, c) \to -f$. Furthermore, as $M \to 0$, $\mu(c(\cdot), M) \to 0$. Then, by Assumption 1(iii) for small $M > 0$, $\Pi(0, c(0)) > 0$ for all $c(\cdot)$. Therefore, $Z(m^o, c; c(\cdot), M) > 0$ for small M. Thus, for all (m^o, c), there exists a unique M such that $Z(m^o, c; c(\cdot), M) = 0$. Therefore, M^x is the unique value such that $Z(m^o, c(m^o); \mu(c(\cdot), M^x)) = 0$ for all $m^o \geq 0$. Continuity of M^x follows from the continuity of Z, and because $Z(m^o, c(m^o); \mu(c(\cdot), M))$ is strictly decreasing in M, M^x is strictly increasing in $c(\cdot)$.

Similarly, following Lemma 7 in Hopenhayn (1992), it can be shown that M^e is continuous and non-increasing on D as long as $V^e(0) > \kappa$. As $M \to \infty$, $V^e(\mu(c(\cdot), M^e)) \to 0$ and as $M \to 0$, $V^e(0) > \kappa$. Because $V^e(\mu(c(\cdot), M))$ is continuous and strictly decreasing in M, there exists M^e such that $V^e(\mu(c(\cdot), M^e)) = \kappa$. M^e is continuous because it is the minimizer of the continuous function $|V^e(\mu(c(\cdot), M)) - \kappa|$. M^e is also non-increasing because $\mu(c(\cdot), M)$ is non-increasing in $c(\cdot)$.

Now note that for any $m^o > 0$, $V(m^o, c; \mu(c(\cdot), M)) > V(0, c; \mu(c(\cdot), M))$ for $c > c^*(m^o)$ because V is strictly increasing in m^o. The fact that $V(m^o, c; \mu(c(\cdot), M))$ has a maximum at $\underline{c}$ then implies that $M^x(c(\cdot)) > M^e(c(\cdot))$ for $c(\cdot) \equiv \underline{c}$. This implies that either there exists some function $c^*(\cdot)$ such that $M^x(c^*(\cdot)) < M^e(c^*(\cdot))$, or $M^x(c(\cdot)) > M^e(c(\cdot))$ for all $c(\cdot) \in D$. In the former case, there exists an equilibrium where $M^x(c^*(\cdot)) = M^e(c^*(\cdot)) > 0$ for some

$c^*(\cdot) \in D$, and in the latter there exists a stationary equilibrium with no entry and exit, i.e. for some $\overline{c} < \infty$ and the exit rule $c^*(\cdot) \equiv \overline{c}$, we have $M^x(c^*(\cdot)) = M^e(c^*(\cdot)) = 0$. Overall, then, there exists a stationary equilibrium.

For positive entry and exit in equilibrium, entry must be relatively easy. If the entry cost satisfies $\kappa < \kappa^* = V^e(\mu^*)$ for some $\kappa^* > 0$, then there exists some $c^*(\cdot) \in D$ such that $M^x(c^*(\cdot)) < M^e(c^*(\cdot))$. Unless the advertising cost and fixed cost is very high, potential entrants have positive expected profit, i.e. $V^e(\mu^*) > 0$, so that $\kappa^* > 0$. Therefore, an equilibrium with positive entry and exit exists.

Uniqueness: Suppose that there are two exit schedules $c_1^*(\cdot)$ and $c_2^*(\cdot)$ such that the corresponding measures μ_1^* and μ_2^* constitute stationary equilibria with positive entry and exit. Assume, without loss of generality, that $c_1^*(0) > c_2^*(0)$, that is, to survive the marginal entrant needs to be more efficient in economy 2 than in economy 1. Then, we must have $V_2^*(0, c_1^*(0)) < V_2^*(0, c_2^*(0)) = 0$ and $V_1^*(0, c_1^*(0)) = 0$. Thus, there must exist some firm type $(0, c)$ such that $V_2^*(0, c) < V_1^*(0, c)$. But free entry requires $V^e(\mu_1^*) = V^e(\mu_2^*) = \kappa$. Therefore, firm value cannot be lower for all entrant types under μ_2^* compared to μ_1^*, and must increase for some. In other words, if the values of all entrant types move in the same direction in response to a change in the measure of firms, the free entry condition is violated under μ_2^* and the two equilibria cannot coexist. We will show that this is the case. Assume, without loss of generality, that when the equilibrium measure of firms μ_1^* changes to μ_2^*, the profit of the most efficient entrant decreases, i.e. $V_2^*(0, \underline{c}) < V_1^*(0, \underline{c})$. If the profits of all entrants with marginal cost higher than $\underline{c}$ also decrease, then the equilibrium is unique. Take any cost level $c' \in (\underline{c}, \min\{c_1^*(0), c_2^*(0)\}]$. Consider the set of firm types that provide a consumer at least as much surplus as firm $(0, c)$ under measure μ_i, $i = 1, 2$

$$B_i(0, c) = \{(x, y) : W^*(x, y) \geq W^*(0, c)\}.$$

Since $V_2^*(0, \underline{c}) < V_1^*(0, \underline{c})$, we must have $\mu_2(B_2(0, \underline{c})) > \mu_1(B_1(0, \underline{c}))$ because profit is strictly decreasing in the measure of firms that offer more consumer surplus than firm type $(0, \underline{c})$. This implies $M_2^* > M_1^*$, i.e. entry mass must be higher for μ_2^*. But since μ^* is strictly increasing in M, we must then also have $\mu_2(B_2(0, c')) > \mu_1(B_1(0, c'))$, and therefore $V_2^*(0, c') < V_1^*(0, c')$ for all $c' \in (\underline{c}, \min\{c_1^*(0), c_2^*(0)\}]$. Thus, we have $V^e(\mu_2^*) < V^e(\mu_1^*) = \kappa$ and the free

entry condition is violated under μ_2^*, implying uniqueness. This completes the proof of Lemma 3 and Theorem 1. ■

Proof of Proposition 5. Part (i) Let F_T^* and f_T^* denote, respectively, the cumulative distribution function and the density of firm size for firms of age $T \geq 1$. We will show that $F_T^*(m) < F_{T-1}^*(m)$ for all $T \geq 1$. Let $c^*(m, m^o) \in [\underline{c}, \infty)$ be the largest cost shock that results in a firm size of at least m at the end of a period, starting from a previous period size of m^o. The function $c^*(m, m^o)$ is decreasing in its first argument and increasing in its second argument. To see this claim, note that $c^*(m, m^o)$ is implicitly defined by the law of motion (7) as

$$m^o(1-\lambda)\widetilde{z}^*(m^o, c^*(m, m^o)) + a^*(m^o, c^*(m, m^o))\widetilde{\omega}^*(m^o, c^*(m, m^o)) = m.$$

Total differentiation with respect to m^o and m gives

$$\begin{aligned}
\frac{\partial c^*(m, m^o)}{\partial m^o} &= -\frac{(1-\lambda) + \frac{\partial a^*}{\partial m^o} + m^o(1-\lambda)\frac{\partial \widetilde{z}^*}{\partial m^o} + a^*\frac{\partial \widetilde{\omega}^*}{\partial m^o}}{\frac{\partial a^*}{\partial c}\widetilde{z}^* + m^o(1-\lambda)\frac{\partial \widetilde{z}^*}{\partial c} + a^*\frac{\partial \widetilde{\omega}^*}{\partial c}} > 0, \\
\frac{\partial c^*(m, m^o)}{\partial m} &= \left(\frac{\partial a^*}{\partial c}\widetilde{z}^* + m^o(1-\lambda)\frac{\partial \widetilde{z}^*}{\partial c} + a^*\frac{\partial \widetilde{\omega}^*}{\partial c}\right)^{-1} < 0,
\end{aligned}$$

where the inequalities follow from the signs of the individual terms established in earlier propositions. The conditional probability that the current period size is at most m can then be written as

$$P^*(m|m^o) = \begin{cases} 0 & \text{if } c^*(m, m^o) < \underline{c}, \\ 1 - H(c^*(m, m^o)) & \text{if } c^*(m, m^o) \in [\underline{c}, c^*(m^o)). \end{cases}$$

Therefore, for age $T = 1$

$$F_1^*(m) = 1 - H(c^*(m, 0)),$$

because new firms start with no customers. For age $T = 2$,

$$F_2^*(m) = \int_{\underline{m}_1^*}^{\overline{m}_1^*} \left[1 - H(c^*(m, m^o))\right] f_1^*(m^o) dm^o = 1 - \int_{\underline{m}_1^*}^{\overline{m}_1^*} H(c^*(m, m^o)) f_1^*(m^o) dm^o,$$

where $\underline{m}_1^* > 0$ and $\overline{m}_1^*$ are the minimum and maximum firm sizes at age $T = 1$. But note that

$$\int_{\underline{m}_1^*}^{\overline{m}_1^*} H(c^*(m, m^o)) f_1^*(m^o) dm^o > H(c^*(m, 0)),$$

because $H(c^*(m,m^o)) > H(c^*(m,0))$ for all $m^o > 0$, as $c^*(m,m^o)$ is strictly increasing in m^o. Thus, $F_2^*(m) < F_1^*(m)$ for all $m \in [0, \overline{m}_2^*]$. For any $T \geq 3$, one can write

$$F_{T-1}^*(m) = 1 - \int_{\underline{m}_{T-2}^*}^{\overline{m}_{T-2}^*} H(c^*(m,m^o)) f_{T-2}^*(m^o) dm^o, \tag{39}$$

$$F_T^*(m) = 1 - \int_{\underline{m}_{T-1}^*}^{\overline{m}_{T-1}^*} H(c^*(m,m^o)) f_{T-1}^*(m^o) dm^o. \tag{40}$$

Note that $\overline{m}_T^* > \overline{m}_{T-1}^*$ and $\underline{m}_T^* = \underline{m}_1^*$ for $T \geq 1$. Now, suppose that $F_{T-1}^*(m) < F_{T-2}^*(m)$. Because $H^*(c^*(m,m^o))$ is strictly increasing in m^o, first order stochastic dominance, together with (39) and (40), implies $F_T^*(m) < F_{T-1}^*(m)$. By induction, it must then hold that $F_T^*(m) < F_{T-1}^*(m)$ for any arbitrary T.

Part (ii) Follows from part (i).

Part (iii) Follows from part (i) and Proposition 1. ■

B Simulation algorithm

We outline the simulation algorithm used for comparative statics. The baseline advertising technology, $\Phi(a) = \chi a^{\varphi}$, generates positive advertising for any $\varphi > 1$. To see this claim, note that Assumption 1(iii) is satisfied because $\lim_{a \to 0} \Phi'(a) = \Phi'(0) = \chi\varphi 0^{\varphi-1} = 0$ for $\varphi > 1$. Therefore, the marginal cost of advertising at $a = 0$, represented by the second term in the l.h.s. of (13), is zero. What remains to be shown is that the marginal return to advertising represented by the first term in the l.h.s. of (13) is positive, and thus exceeds the marginal cost at $a = 0$. A feasible, but not necessarily optimal, action available to any firm is to charge monopoly price and sell, at least, to its captive consumers. This action yields positive revenue per ad for a firm, because, as long as advertising is not free, there is always a positive mass of consumers whose only ad is the ad they received from the firm. Formally, the revenue per ad for a type (m^o, c) firm when it charges its monopoly price $p^m(m^o, c) \geq s$ is

$$\omega^*(p^m, m^o)(p^m - c + R^*) \geq \Omega^*(0)(p^m - c + R^*) > \Omega^*(0)(p^m - c) > \Omega^*(0)(s - c),$$

where the first inequality follows from the fact that $\omega^*(p^m, m^o) \geq \Omega^*(0)$, the second from $R^* \geq 0$, and the third from $p^m \geq s$. Because $\Omega^*(0) > 0$, the last term, $\Omega^*(0)(s-c)$, is positive

as long as $s > c$, which is satisfied for all $c < 4$ because $s = 4$ and $c \sim U[1, 5]$. Consequently, there is positive advertising as long as the fixed cost is not too high and the industry is not empty, i.e. the entry cost is not too high. For the industry to be non-empty, in a given period there must be some firm types which do not exit, and the expected value of entry must not be lower than the entry cost. The conditions for a non-empty industry are checked as part of the simulation algorithm below.

The algorithm is based on Theorem 1 and involves nested value function iterations. The simulation setup requires a grid of firm types. We partition the support of the marginal cost distribution $U[1, 5]$ into a discrete set $\{c_1, ..., c_n\}$, where $c_1 = 1$ and $c_n = 5$. We also use a grid of firm size levels $\{m_1, ..., m_k\}$, where $m_1 = 0$ and $m_k = \overline{m}$, where $\overline{m}$ is taken sufficiently large to contain the upper bound on firm size in stationary equilibrium. The algorithm has the following steps:

1. Start with initial guesses for the maximum price charged, $p_m \geq s$, the value function, $V(m^o, c)$, the surplus function, $W(p, m^o)$, the measure of firms, $\mu(m^o, c)$, and the cumulative distribution of ads, $Q(p, m^o)$ (or equivalently advertising policy function $a(m^o, c)$). We look for fixed points by simultaneously iterating on these variables and functions.[1] As initial values, we choose $p_m = 4.5$, $V(m^o, c) \equiv \alpha_1 m^o + \alpha_2 c$ for some $\alpha_1 \geq 0$, and $\alpha_2 \leq 0$, $W(p, m^o) \equiv 0$, $W^{\varnothing} = 0$, and $\Delta\mu(m^o, c) = b \geq 0$ such that $\sum_{m^o} \sum_c \Delta\mu(m^o, c) = \sum_{m^o} \sum_c b = 1$, implying that $b = \frac{1}{nk}$. $Q(p, m^o)$ is also set to be the $c.d.f.$ of a two-dimensional discrete uniform distribution over $\{p_1, ..., p_l\} \times \{m_1, ..., m_k\}$. Given $W(p, m^o)$ and $Q(p, m^o)$, compute the individual demand functions as

$$\begin{aligned} z(p, m^o) &= \sum_{i=0}^{T} \frac{e^{-A} A^i}{i!} Q^i(p, m^o), \qquad (41) \\ \omega(p, m^o) &= Q_s(p, m^o) \sum_{i=0}^{T} \frac{e^{-A} A^i}{i!} Q^i(p, m^o) \end{aligned}$$

[1] There are two alternative methods of iteration on these functions. One method is to construct separate loops of iteration for each function. The second method is simultaneous iteration of all or a subset of the functions under one single loop. The first method is more robust, but computationally slower. The second method is faster with potential problems of robustness. To deal with the robustness issues, small step sizes for updating functions may be used. We tried both methods, and given the dimensionality of the setup, we found that using the second method with small step sizes is easier.

where $Q_s(p, m^o)$ is the probability that among all the sales that take place, the firm type that makes a randomly selected sale is not preferred to firm type (p, m^o). Since we do not have $Q_s(p, m^o)$ available initially, we start by assuming that $Q_s(p, m^o) = Q(p, m^o)$, and update it in further iterations as a computed $Q_s(p, m^o)$ becomes available. The upper limit T in the summations (41) is set to a large enough number ($T = 20$) and the term $\frac{e^{-A}A^i}{i!}$ follows from a Poisson approximation to $\Psi(i)$.

2a. Compute $\frac{\Delta z(p_i,m^o)}{\Delta p_i} = \frac{z(p_i,m^o)-z(p_{i-1},m^o)}{p_i-p_{i-1}}$, $\frac{\Delta \omega(p_i,m^o)}{\Delta p_i} = \frac{\omega(p_i,m^o)-\omega(p_{i-1},m^o)}{p_i-p_{i-1}}$, and $\frac{\Delta V(m_j^o,c)}{\Delta m_j^o} = \frac{V(m_j^o,c)-V(m_{j-1}^o,c)}{m_j^o-m_{j-1}^o}$ over a grid for p_{kn} and m_n^o. Solve the discrete-space version of the corresponding first order conditions (11) and (13), and find the optimal price and advertising levels $a_{i,j}$ and $p_{i,j}$ for each firm type using

$$p_{ij} = c_j - \frac{m_i^o(1-\lambda)z(p_{ij}, m_i^o) + a_{ij}\omega(p_{ij}, m_i^o)}{m_i^o(1-\lambda)\frac{\Delta z(p_{ij},m_i^o)}{\Delta p_{ij}} + a_{ij}\frac{\Delta \omega(p_{ij},m_i^o)}{\Delta p_{ij}}} - \beta \sum_k \frac{\Delta V(m^*, c_k')}{\Delta m^*} h(c_k'). \tag{42}$$

where $m^* = m_i^o(1-\lambda)z(p_{ij}, m_i^o) + a_{ij}\omega(p_{ij}, m_i^o)$, and

$$a_{ij} = \left[\frac{\omega(p_{ij}, m_i^o)\left[(p_{ij} - c_j) + \beta \sum_k \frac{\Delta V^o(m^*,c_k')}{\Delta m^*} h(c_k')\right]}{\varphi \chi} \right]^{\frac{1}{\varphi - 1}}. \tag{43}$$

2b. Compute period sales and profits for all firm types

$$m_{ij}^* = m_i^o(1-\lambda)z(p_{ij}, m_i^o) + a_{ij}\omega(p_{ij}, m_i^o),$$

$$\Pi(m_i^o, c_j) = m_{ij}^*(p_{ij} - c_j) - \Phi(a_{ij}) - f.$$

Compute total number of ads

$$A = \sum_i \sum_j a_{ij} \Delta \mu(m_i^o, c_j).$$

Obtain the updated value function

$$V'(m_i^o, c_j) = \max\{0, \Pi(m_i^o, c_j) + \beta \sum_{k=1}^{n} V(m_{ij}^*, c_k) h(c_k)\}.$$

Compute the exit indicator

$$X(p_{ij}, c_j) = \begin{cases} 0, & \text{if } \Pi(m_i^o, c_j) + \beta \sum_{k=1}^n V(m_{ij}^*, c_k) h(c_k) \geq 0, \\ 1, & \text{otherwise.} \end{cases}$$

2c. Obtain the updated surplus functions for consumers,

$$W^{\varnothing\prime} = \frac{\beta(1-\lambda)}{(1-\beta(1-\lambda)e^{-A})} \sum_{i=1}^{T} \frac{e^{-A}A^i}{i!} \sum_k W_k \Delta G^i(W_k)$$

$$\begin{aligned} W'(p_{ij}, m_j^o) &= s - p_{ij} \\ &+\beta(1-\lambda)\left\{ e^{-A}[(1-X(p_{ij},m_j^o))\sum_k W(p'_{kj}, m^*_{ij})]h(c'_k) + X(p_{ij},m_j^o)W^{\varnothing} \right. \\ &+\sum_{i=1}^{T} \frac{e^{-A}A^i}{i!} \left[(1-X(p_{ij},m_j^o)) \sum_k \max\{W_k, W(p'_{kj}, m^*_{ij})\} \Delta G^i(W_s) \right. \\ &\left.\left. +\ X(p_{ij},m_j^o)) \sum_k W_k \Delta G^i(W_k) \right] \right\}. \end{aligned} \tag{44}$$

2d. Obtain the implied distribution of ads across firm types $Q'(p_{ij}, m_i^o)$ using the optimal number of ads a_{ij} sent by each firm type and the measure of firms $\mu(m_i, c_j)$, i.e.

$$Q'(p_{ij}, m_i^o) = \frac{1}{A} \sum_{(q,r) \text{ s.t. } W(p_{qr},m_q) \leq W(p_{ij},m_i)} a_{qr} \Delta\mu(m_q^o, c_r).$$

Also, obtain the implied distribution of sales across firm types $Q'_s(p_{ij}, m_i^o)$

$$R = \sum_i \sum_j m^*_{ij} \Delta\mu(m_i^o, c_j),$$

$$Q'_s(p_{ij}, m_i^o) = \frac{1}{R} \sum_{(q,r) \text{ s.t. } W(p_{qr},m_q^o) \leq W(p_{ij},m_i^o)} m^*_{qr} \Delta\mu(m_q^o, c_r).$$

3. Calculate the distances

$$\begin{aligned} d(Q, Q') &= \max_{i,j} |Q(p_{ij}, m_i^o) - Q'(p_{ij}, m_i^o)|, \\ d(V, V') &= \max_{i,j} |V(m_i^o, c_j) - V'(m_i^o, c_j)|, \\ d(W, W') &= \max_{i,j} |W(p_{ij}, m_i^o) - W'(p_{ij}, m_i^o)|. \end{aligned}$$

If $d(Q, Q') < \varepsilon$, $d(V, V') < \eta$, $d(W, W') < \sigma$ for small $\varepsilon, \eta, \sigma > 0$, then go to step 4. Otherwise, set $V \equiv \alpha V + (1-\alpha)V'$, $W \equiv \alpha W + (1-\alpha)W'$, and $Q \equiv \alpha Q + (1-\alpha)Q'$ (with $\alpha = 0.1$) and go back to Step 2.

4. Obtain the entry mass M using the discrete version of the free entry condition,

$$M = \sup\{M : \sum_{i=1}^{n} V'(0, c_i)h(c_i) = \kappa\},$$

and compute the exit rule $c(m^o)$ point-wise as

$$c(m_i^o) = \max\{c_j : V'(m_i^o, c_j) > 0 \text{ and } V'(m_i^o, c_{j+1}) \leq 0\}.$$

5. Using M, and $c(m_i^o)$, and $\mu(m_i^o, c_j)$, obtain the updated measure

$$\mu'(m_i^o, c_j) = \sum_{m \leq m_i^o} \sum_{c' \leq \min\{c_j, c^o(m^o)\}} \mu(m, c') + M \sum_{c' \leq \min\{c_j, c^o(0)\}} h(c').$$

6. Calculate the distance

$$d(\mu, \mu') = \max_{(i,j)} |\mu(m_i^o, c_j) - \mu'(m_i^o, c_j)|.$$

If $d(\mu, \mu') < \xi$ for some small $\xi > 0$, stop. If $M > 0$ and $c(m^o) > c_1$, equilibrium entails positive entry and exit. Otherwise, set $\mu(m^o, c) = \alpha\mu(m^o, c) + (1 - \alpha)\mu'(m^o, c)$, $V \equiv \alpha V + (1 - \alpha)V'$, $W \equiv \alpha W + (1 - \alpha)W'$, $Q \equiv \alpha Q + (1 - \alpha)Q'$ (with $\alpha = 0.1$) and go back to Step 2.

C Calculation of Total Factor Productivity

Plant (establishment) level TFP is aggregated to the firm level by weighting each plant's TFP measure in a given industry by its share of total value of firm's shipments in that industry. Recognizing that many plants manufacture several products that fall within more than just one 4-digit SIC industry, we use the multifactor superlative index number for the revenue-based productivity measurement. This index measures a plant's productivity relative to other plants in its main 4-digit SIC industry – the industry in which the plant has the largest value of shipments. For details on this index, see the discussion of Malmquist productivity indices in Section 4 of Caves, Christensen and Diewert (1982). A similar approach is followed by Bernard, Redding and Schott (2010). An establishment's output is measured using deflated (real) value of shipments plus nominal inventory investments. The inputs are measured using deflated value of equipment, deflated value of plant, total labor

hours, deflated value of materials, and deflated value of energy. Labor inputs are measured as an establishment's production-worker hours adjusted by multiplying the production-worker hours by the ratio of total payroll to payroll of production workers. Equipment and plant inputs are establishment's book values for their structure and equipment stocks deflated using sector-specific deflators from the Bureau of Economic Analysis. Materials and energy inputs are based on establishment's expenditures on materials and energy deflated using the input price indices in NBER Productivity Database. For the calculation of 4-digit industry-level cost shares, materials and energy expenditures and payments to labor are aggregated across establishments in the industry. Industry level cost of capital is obtained by first multiplying an establishment's real capital stock with the capital rental rates for the 2-digit industry an establishment belongs to, and then aggregating across establishments in the 4-digit industry. Each input cost is divided by the total cost at the industry level to obtain the cost share of the input. The quantity-based (physical) TFP is calculated in the same way as in Appendix A2 of Foster, Haltiwanger and Syverson (2008). Because the firms in the sample for price regressions are highly specialized in their primary products and these products are highly homogenous, we do not use a multi-product productivity measurement for the case of physical TFP.

D Industry Definitions

A constraint on the selection of the industries in Table 3 was the switch from SIC codes to NAICS codes in 1997, and a further revision in 2002. These changes coincided with a period during which Internet-based commerce diffused rapidly, rendering construction of consistent time series of industry aggregates difficult for other candidate industries. The 1987 SIC (which also applies to 1992), 1997 NAICS and 2002 NAICS codes are comparable for all industries except for Life Insurance. For Life Insurance, the discrepancy due to code revisions is small, as explained below. The following information about the industries analyzed in Table 3 is taken from US Census Bureau's Industry Statistics Sampler.

New Car Dealers (SIC 5511 / NAICS 441110) Establishments primarily engaged in retailing new automobiles and light trucks, such as sport utility vehicles, and passenger and

cargo vans, or retailing these new vehicles in combination with activities, such as repair services, retailing used cars, and selling replacement parts and accessories.

Book Stores (SIC 5942/NAICS 451211) Establishments primarily engaged in retailing new books.

Camera and Photographic Supplies Stores (SIC 5946/NAICS 443130) Establishments primarily engaged in either retailing new cameras, photographic equipment, and photographic supplies or retailing new cameras and photographic equipment in combination with activities, such as repair services and film developing.

Prerecorded Tape, Compact Disc, and Record Stores (SIC 5735/NAICS 451220) Establishments primarily engaged in retailing new prerecorded audio and video tapes, compact discs (CDs), digital video discs (DVDs), and phonograph records.

Travel Agencies (SIC 4724/NAICS 561510) Establishments primarily engaged in acting as agents in selling travel, tour, and accommodation services to the general public and commercial clients.

Life Insurance (SIC 6311/NAICS 524113 and 524130) Establishments primarily engaged in underwriting life insurance. These establishments are operated by enterprises that may be owned by stockholders, policyholders, or other carriers. The sales classified under SIC 3611 are made up of 95% of the sales classified under NAICS 524113 plus nearly all of the sales classified under NAICS 524130. However, NAICS 524130 contains only 2.2% of the total establishments in SIC 6311 and have a small contribution to total sales. Therefore, we exclude NAICS 524130 in calculating the industry statistics for 1997 and 2002.

E Additional Tables and Figures

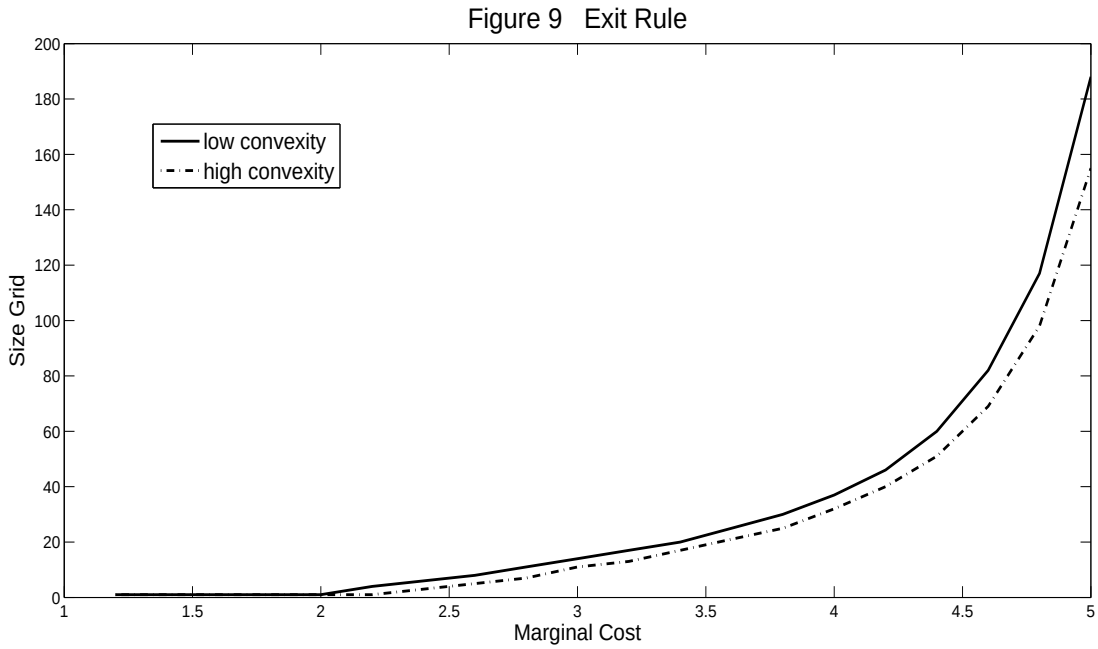


Figure 9: Convexity of information cost: Exit rules

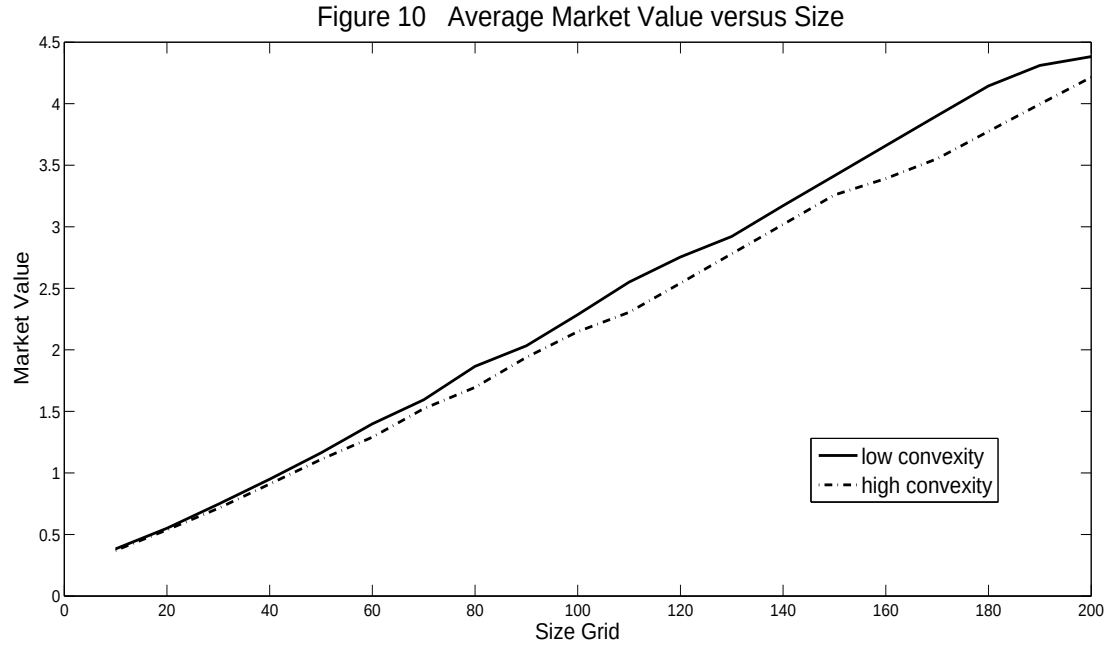


Figure 10: Convexity of information cost: Firm size-value relationship ("0"=smallest)

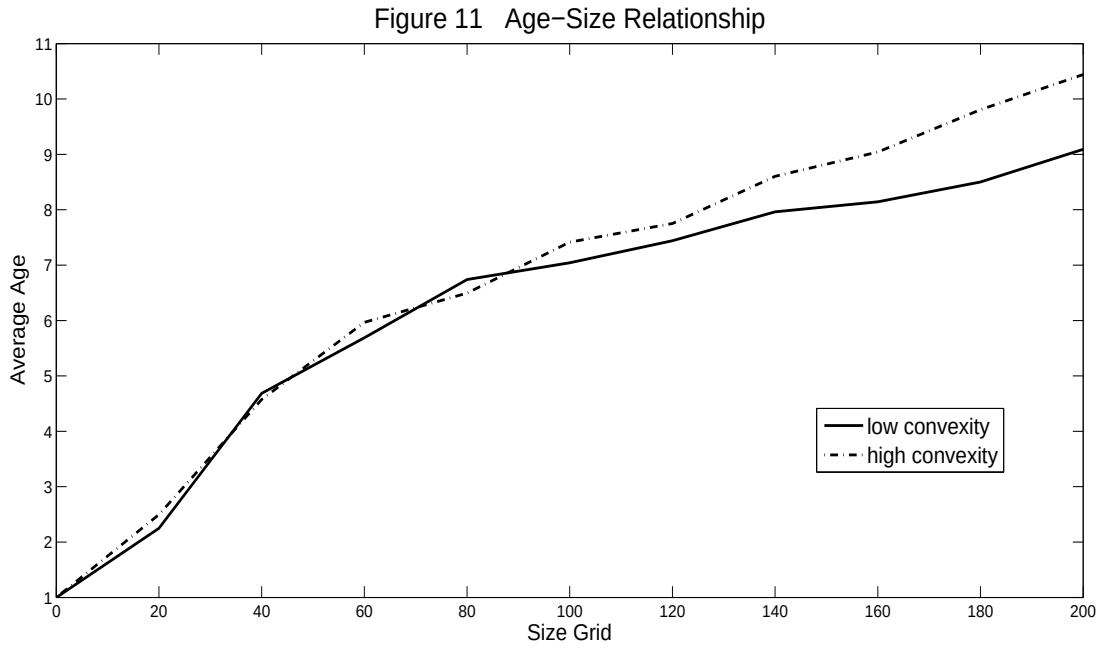


Figure 11: Convexity of information cost: Firm age-size relationship ("0"=smallest)

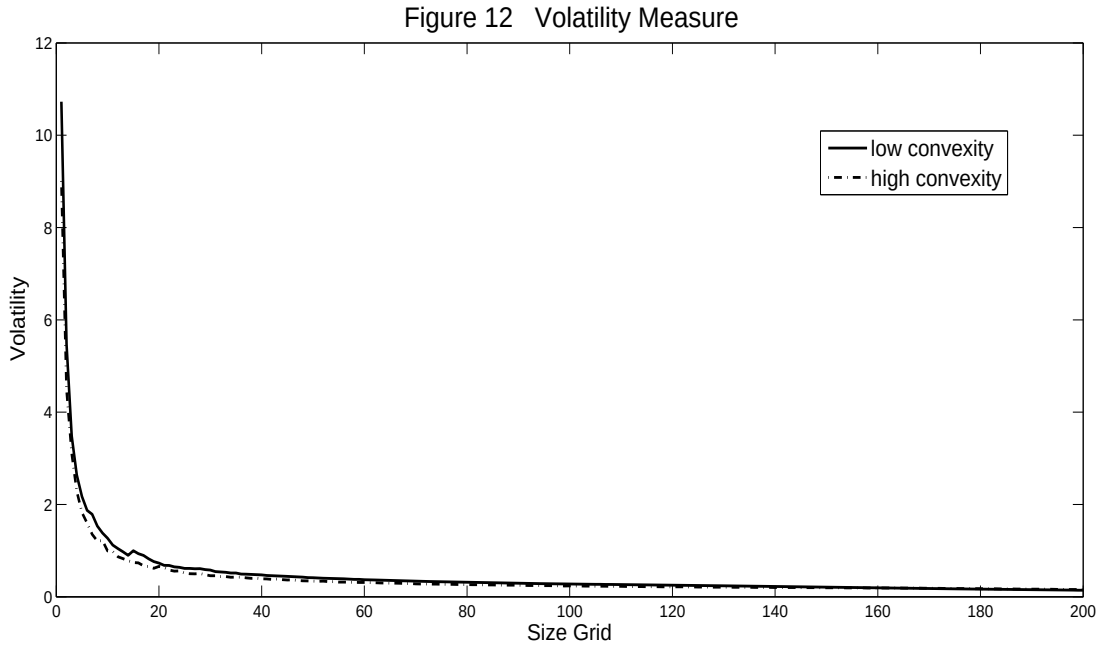


Figure 12: Convexity of information cost: Firm size-volatility relationship ($Vol(m^o)$) ("0"=smallest)

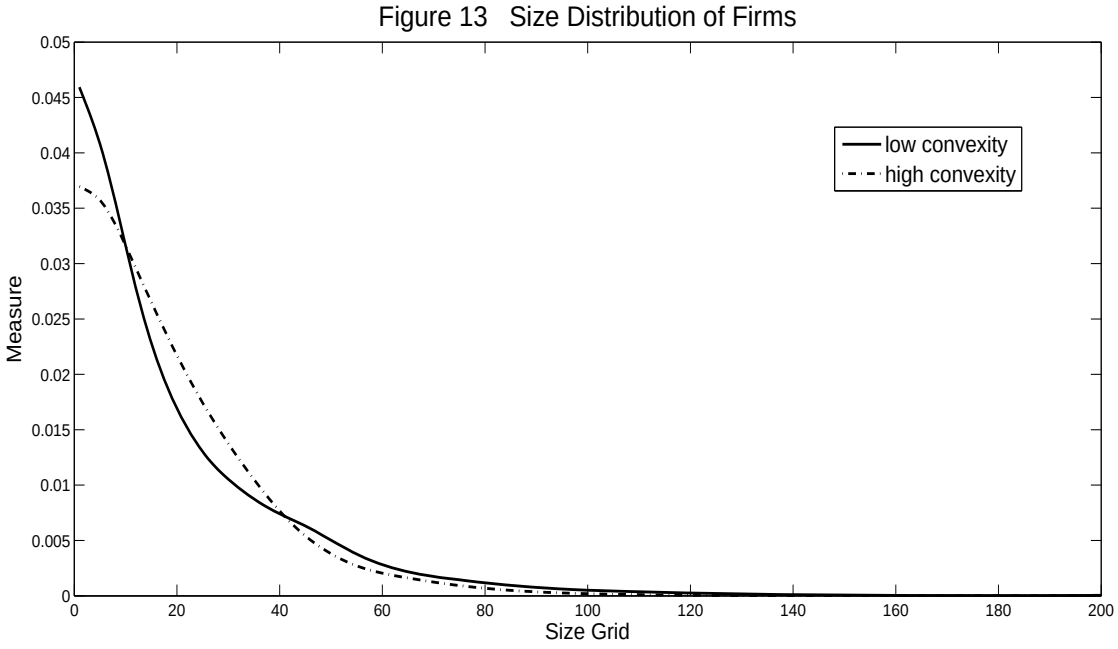


Figure 13: Convexity of information cost: Density of firm size

	Dependent Variable: 1997 Advertising Expenditures ($K)			
	IV		V	
Independent variables:	OLS	Tobit	OLS	Tobit
1996 Value of Shipments ($K)	0.012***	0.013***	-	-
	[0.003]	[0.001]		
1996 Number of Employees	-	-	4.543***	5.113***
			[1.007]	[0.084]
1997 TFP	0.194**	1.341**	0.155**	1.298**
	[0.045]	[0.402]	[0.041]	[0.403]
Multi-unit dummy	334.9	5,771.1***	175.3	5,234.1***
	[298.9]	[345.2]	[367.8]	[276.4]
Industry fixed effects	Y	Y	Y	Y
Inverse Mills Ratio (IMR)	0.149***	-	0.147***	-
	[0.041]		[0.042]	
R^2 / Log Likelihood	0.36	-201,878	0.28	-202,345

Notes: All variables are in levels. OLS estimation is based on Heckman's two-step correction. IMR is the ratio of the standard normal density to the standard normal c.d.f. predicted from the selection equation.
Coefficients in OLS estimation (except for that of IMR) are marginal effects that take selection into account.
Corrected standard errors in parentheses. (*), (**), and (***) indicate significance at 10%, 5%, and 1% respectively.
Industry fixed effects are based on 1997 SIC codes (4-digit).

Table A1. The relationship between firm advertising expenditures, size, and productivity (ASM Sample)